

B.SC. ENGG. PROJECT

LERNEN

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(In partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science & Engineering)



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November 26, 2021

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Chapter 1

INTRODUCTION

1.1 Introduction

Lernen is a web based application which will make easier to find learning materials of various instructor. Lernen is all about connecting discovering talents, inspiring people and impact lives with the best teaching and earning opportunities. “Lernen” is an online teaching platform, where instructors can create and upload their knowledge. They also can upload their books PDF. It is an E-Learning Platform to learn anything from anywhere from the best teachers. This project is mainly based on e-learning system. E-Learning is a vast and expanding platform with a huge perspective in education. we can also say that, e-learning is time and cost saving learning procedure. there are mainly 4 factors to make our project effective, such as,

- learner
- learning materials
- learning atmosphere
- technology

It bridges the gap between a teacher and a student in different ways.

The aim behind this project is to provide a user-friendly environment for imparting knowledge and giving everyone the opportunity to learn at a low cost, no matter where they are. The specific objectives can be summarized in combining electronic content management systems,

help students to exchange ideas which help to think creatively, accessibility of educational materials at low cost, encourage students to achieve their goals. The students will be able to register online, read the course material and give online exam to judge themselves.

1.2 Project overview

The website is reduced as much as possible to avoid errors while entering the data. It also provides an error message while entering invalid data. No formal knowledge is needed for the user to use this system. Thus by this all it proves it is user-friendly. This e-learning Site System, as described above, can lead to error free, secure, reliable and fast management systems. It can assist the user to concentrate on their other activities rather than concentrating on the record keeping. Thus it will help organizations in better utilization of resources.

1.3 Problem Statement

The current situation is very limited to few resources, students are unable to get knowledge more than that the lecture provides to them. This in the end limits student's performances, because everything a student gets is collected from lectures in class. Here are some of the problems of the current system:

- The current system is that only owner upload lecture tutorial for students.
- Students submit assignment to lectures through hard copies or personal emails.
- Students only get help if the lecturer are free.
- New lectures to a course have to get materials on their own.
- Student are required to physical be in the classroom in order to gain knowledge thereby sacrificing all other responsibilities.

1.4 Motivation

Facing the problems described above, we are willing to build a custom system called Lernen (E-Learning System) that can solve the following problems :

- Instructors of various field can upload their course
- Students from anywhere can found their preferred course in one place
- Preparing properly for their academic courses
- Sharing valuable information about any topic accurately
- Can discuss with one another that will increase their knowledge
- Get resources without any difficulties
- Not getting enough response of questions about a topic
- Distributing knowledge easily to the learners

1.5 Objectives

- Provide a user-friendly environment for imparting knowledge and giving everyone the opportunity to learn at a low cost.
- Provide inexpensive, efficient and comfortable way for learning.
- Help students to exchange ideas, which help to think creatively.
- Accessibility of educational materials at low cost.
- provide distance learning.
- Computer evaluated tests.
- Reduced paper work.

1.6 Development Tools

Lernen is a website of e-learning system, integrated with XAMPP. To build this website basically we use PHP, HTML, CSS and JAVASCRIPT. HTML(Highper text Markup Language) and CSS(Cascading Style Sheet) For layout,JAVASCRIPT for animation and transformation, and PHP Used for coding. We use Xampp for MySQL database. You can give default attributes values for different elements in HTML, which will be used(for that particular element) in website until you change that attribute in corresponding JAVASCRIPT file for that particular element. To change attributes one must first define an id to the element and use that id in JAVASCRIPT file to change it's attributes.

- Technology
 - MySQL for database management
 - Different package managers for different purpose
 - XAMPP for running PHP on the server-side machine
- Language
 - PHP
 - HTML
 - CSS
 - JAVASCRIPT
 - SQL

1.7 Proposed System

In our proposed system there will be two actors. They are students and instructor. users will register themselves in the registration page by giving email name, password. After giving this information, an otp will be sent to their email to confirm the registration.

Here instructor will Add courses along with its video, slide and various level quizzes.They also

can add books along with it's pdf. Instructors can change their course video if necessary. They can add or delete slide image and quiz questions.

The students will provide feedback by giving ratings and comments on a particular course. Average rating and user comments will be shown in course details page. After seeing video and slide students can give 3 level quiz tests to judge themselves. A summery will show after the quiz test by whih students can identify their mistakes. They can find their registered course in mycourse section. After completing a course they can archive that course to avoid mess. In history section, they can see their course status and registration date.

From our website users can download books or read books without download.

From profile users can update their information. Users can search for the course of their choice.

1.8 Purpose Of The System

e-Learning is vast and an expanding platform with huge prospective in advanced education. Since there are numerous challenges in making e-Learning effective, it's important to know howto manage it and access to the coffers. Take a nanosecond and just imagine if one isn't havingthe road map to guide from launch to finish is actually like plunging into e-Learning withoutan effective strategy because learners would be lost in the literacy content.

One of the ideas behind the Lernen is to encourage students to learn, no matter where they are. The physical classroom learning nowadays is no longer applicable for the current younger generations. Internet and distance learning which is generally known as online education plays a vital roles in the country's education system. It is undeniable that online education provides ample of benefits to young learners. Also many skilled people cannot use their skills due to lack of opportunities and resources. Lernen will provide opportunities for them to improve their skills and use their talents to achieve their goals.

1.9 Goal

The Major Goals Of this project is to provide a user-friendly environment to provide knowledge and give everyone a chance to learn, irrespective of where they are. Another goal is enhance the quality of learning and teaching, Improve the efficiency and effectiveness of learning Improve user-accessibility and time flexibility to interact learners within the learning process.

Chapter 2

EXISTING SYSTEM

2.1 Introduction

E-learning is among the most important explosion propelled by the internet transformation. Although it has the inability to handle all functions of the institution such as some courses that require practical skills and supervision but it also increases the interaction among students and lectures which in turn will lead to achieve the learning goal as students are able to access anywhere and anytime (Noeline, 2010). E-learning delivers content through electronic information and communications technologies (ICTs). According to, the use of these facilities, involves various methods which includes systematized feedback system, computer-based operation network, video conferencing and audio conferencing, internet worldwide websites and computer assisted instruction. This delivery method increases the possibilities for how, where and when employees can engage in lifelong learning. Therefore an E-learning system has more advantages than it has limitations.

There are two main aims to this existing system analysis. The first aim is to establish the characteristics and importance of formative, coursework assessment. The second aim is to identify e-learning techniques, tools and approaches for this type of assessment, to discuss what is known about their effectiveness and to uncover factors influencing uptake. For the first aim key articles and books that have shaped current theory on formative assessment of student work have been identified and analyzed. For the second aim a comprehensive search of e-learning literature from 2001 to the present was conducted.

2.2 Existing System

We have visited several systems. They are listed below.

2.2.1 10 minute school

10 minutes school is the largest online education platform of Bangladesh. It was introduced by Ayman Sadik with his goal to destroy all kinds of barriers of ensuring quality education to all across Bangladesh. Basically 10 minute school teaches everything starting from class 1 to university admission test related all kinds of education all across over Bangladesh throughout the 64 districts and has a reach of 2.5 million students in everyday.

Advantages

- It connects students from any part of the world, saves time money and provides opportunities for self-learning and development at the students' own pace.
- Easy to start. Low price and availability of popular courses is the main advantage of this platform.
- 10 Minute School offers all these benefits by making education a fun experience accessible to all.
- 10 minute school has interactive Smart books on all academic subjects starting from English to General knowledge and many other subjects where students can learn complex subjects in details through interactive books created to allow students to hover over different aspects of the content and know the in-depth meaning associated with it.
- 10 minute School has the largest online community of learners with four Facebook Groups catering to academics , English language learning, skill development and BBA-MBA grooming.
- They offer some free courses.
- They also provide Books on various subjects.
- while attending live classes, the students can win exciting prizes like 4G handsets, smart watches, headphones and cricket bats signed by national cricket team players.

2.2.2 Instructory

Instructory is the Online teaching marketplace in Bangladesh to earn money. And best E-Learning platform to learn online courses from the top instructors.

Advantages

- A teacher can easily earn by sharing their classes with everyone and students can learn from online video tutorials on various topics.
- It has created earning opportunities for millions of instructors!
- It is an E-Learning Platform to learn anything from anywhere from the best Top rated Professionals and Freelancers!
- Course prize are reasonable.
- Payment is also easy: one can opt for bKash, Visa, MasterCard, mobile or internet banking. And as an instructor, you get your share by 20-25th of every month.
- Course payment can be done by Cards or Mobile Banking Or Net Banking.
- Although the idea of Instructory originated from academic demands, it is also hosting content on life and professional skills i.e. programming, website development, photography, graphics design, SEO and digital marketing.

Disadvantages

- They don't offer books.
- No free Course is available.

2.2.3 Ghuri Learning

Ghoori Learning is very reputed and trusted Online Education Websites in Bangladesh where you can learn various courses and can benefit from the contents of Ghoori Learning. In this online platform there are various course for different level of students. Moreover, students can watch chapter and topic-wise pre-recorded lecture videos, attempt quiz.

Advantages

- It provides Udemy gift card, BBC Janala online course.
- Ghuri Learning is offering many new educational and skill development courses.
- They are offering courses for IELTS and various exam preparation (BCS, University Admission, SSC and HSC). They have courses in digital marketing, photography, language learning (Arabic, Japanese, Chinese and Korean), graphics design, web design and more.
- Course prize are reasonable.
- Students can pay easily by bKash, Visa or MasterCard.
- They offer courses in Bengali.

Disadvantages

- Ghouri Learning is not world wide. They offering courses in only Bengali. So, Only Bangladeshi can attend this course.
- Ghouri Learning is Concentric. One can't upload his course in this site.
- They don't offer books or practice test.
- No free Course is available.

2.3 Supporting Literature

2.3.1 Used Diagram

Entity Relationship Diagram

An Entity-relationship model (ER model) describes the structure of a database with the help of a diagram, which is known as Entity Relationship Diagram (ER Diagram). An ER model is a design or blueprint of a database that can later be implemented as a database. The main components of E-R model are:

- Entity: An entity set is a group of similar entities and these entities can have attributes. In terms of DBMS, an entity is a table or attribute of a table in database. An ER diagram

shows the relationship among entity sets. So by showing relationship among tables and their attributes, ER diagram shows the complete logical structure of a database.

In Our ER diagram we have 14 entities :

- User
 - Course
 - Books
 - Category
 - Registration
 - Video
 - Quiz Questions
 - Quiz Answers
 - Review
 - Rating
 - PPT
 - QA
 - Archive
 - Result
- Weak entity: A weak entity is one that can only exist when owned by another one. The entity sets which do not have sufficient attributes to form a primary key are known as weak entity sets and the entity sets
 - Attribute: Attributes are the properties which define the entity type.
 - Multi valued attribute: An attribute that can hold multiple values is known as multivalued attribute. It is represented with double ovals in an ER Diagram.
 - Derived attribute: A derived attribute is one whose value is dynamic and derived from another attribute. It is represented by dashed oval in an ER Diagram.

- Relationship: A relationship is represented by diamond shape in ER diagram, it shows the relationship among entities.

Mapping Cardinality of ERD

- One-to-One : In a one-to-one relationship, one record in a table is associated with one and only one record in another table.
- One-to-Many: In relational databases, a one-to-many relationship occurs when a parent record in one table can potentially reference several child records in another table. The opposite of a one-to-many relationship is a many-to-many relationship, in which a child record can link back to several parent records.
- Many-to-One: Many-to-One relationship in DBMS is a relationship between more than one instances of an entity with one instance of another entity.
- Many-to-Many: A many-to-many relationship is a type of cardinality that refers to the relationship between two entities, say, A and B, where A may contain a parent instance for which there are many children in B and vice versa.

Data Flow Diagram

A data-flow diagram is a way of representing a flow of data through a process or a system. The DFD also provides information about the outputs and inputs of each entity and the process itself. A data-flow diagram has no control flow — there are no decision rules and no loops. Components of DFD DFDs only involve four symbols. They are:

- Process: An activity that changes or transforms data flows. Since they transform incoming data to outgoing data, all processes must have inputs and outputs on a DFD. This symbol is given a simple name based on its function, such as “Ship Order,” rather than being labeled “process” on a diagram. In Gane-Sarson notation, a rectangular box is used and may be labeled with a reference number, location of where in the system the process occurs and a short title that describes its function. Processes are typically oriented from top to bottom and left to right on a data flow diagram.

- **Data Flow** : Movement of data between external entities, processes and data stores is represented with an arrow symbol, which indicates the direction of flow. This data could be electronic, written or verbal. Input and output data flows are labeled based on the type of data or its associated process or data store, and this name is written alongside the arrow.
- **Data Store** : A data store does not generate any operations but simply holds data for later access. Data stores could consist of files held long term or a batch of documents stored briefly while they wait to be processed. Input flows to a data store include information or operations that change the stored data. Output flows would be data retrieved from the store.
- **External entity** : Also known as actors, sources or sinks, and terminators, external entities produce and consume data that flows between the entity and the system being diagrammed. These data flows are the inputs and outputs of the DFD. Since they are external to the system being analyzed, these entities are typically placed at the boundaries of the diagram. They can represent another system or indicate a subsystem.

Data flow diagrams are also categorized by level. Starting with the most basic, level 0, DFDs get increasingly complex as the level increases.

0 Level Data Flow Diagram

also known as context diagrams, are the most basic data flow diagrams. They provide a broad view that is easily digestible but offers little detail. Level 0 data flow diagrams show a single process node and its connections to external entities.

Level-1 data flow diagram

Level 1 DFDs are still a general overview, but they go into more detail than a context diagram. In a level 1 data flow diagram, the single process node from the context diagram is broken down into sub processes. As these processes are added, the diagram will need additional data flows and data stores to link them together.

Use Case Diagram

Use Case Diagram A use case diagram is a way to summarize details of a system and the users within that system. It is generally shown as a graphic depiction of interactions among different elements in a system. Use case diagrams will specify the events in a system and how those events flow, however, use case diagram does not describe how those events are implemented. A use case is a methodology used in system analysis to identify, clarify, and organize system requirements. In this context, the term "system" refers to something being developed or operated, such as a mail-order product sales and service Web site. Use case diagrams are employed in UML (Unified Modeling Language), a standard notation for the modeling of real-world objects and systems. There are a number of benefits with having a use case diagram over similar diagrams such as flowcharts.

Basic Use Case Diagram Symbols and Notations

- System: A specific sequence of actions and interactions between actors and the system. A system may also be referred to as a scenario.
- Use Case : Horizontally shaped ovals that represent the different uses that a user might have.
- Actors : The users that interact with a system. An actor can be a person, an organization, or an outside system that interacts with your application or system. They must be external objects that produce or consume data.
- Goals: The end result of most use cases. A successful diagram should describe the activities and variants used to reach the goal.

2.3.2 Technology Used

Lernen is a website of e-learning system, integrated with XAMPP. To build this website basically we use PHP, HTML, CSS and JAVASCRIPT. HTML(Higher text Markup Language) and CSS(Cascading Style Sheet) For layout, JAVASCRIPT for animation and transformation,

and PHP Used for coding. We use Xampp for MySQL database. You can give default attributes values for different elements in HTML, which will be used(for that particular element) in website until you change that attribute in corresponding JAVASCRIPT file for that particular element. To change attributes one must first define an id to the element and use that id in JAVASCRIPT file to change it's attributes.

- Technology
 - MySQL for database management
 - Different package managers for different purpose
 - XAMPP for running PHP on the server-side machine
- Language
 - PHP
 - HTML
 - CSS
 - JAVASCRIPT
 - SQL

2.4 Conclusion

E-learning is becoming increasingly prominent in tertiary education. All available evidence points toward growing enrolments and provision albeit from a low starting point. However, after the hype of the new economy, growing disenchantment with e-learning has replaced over-enthusiasm. Failures of e-learning operations have, at least temporarily, over shadowed the prospects of widened and flexible access to tertiary education, pedagogic innovation, and decreased cost that was once embodied by e-learning.

Chapter 3

SYSTEM EVALUATION

3.1 Introduction

Research works in the arena of E-Learning are characterized by a broad diapason of operations, ranged from virtual classrooms to remote courses or distance literacy. Because of a lot of limitations using web2.0 for creating E-learning operation system, now-a-days we use Web3.0 which is known as Semantic web that recovers the limitations of Web2.0. In this paper, we've proposed a model of "E-learning Operation system using Semantic web technologies", where course syllabus, tutoring styles, learning conditioning and learning styles are included and it'll be more applicable in different literacy institutions. We hope that it'll give a inordinate feedback from both preceptors and scholars.

3.2 Requirement Analysis

E-learning has evolved over the last two decades from being a supplementary tool to getting an essential companion to class tutoring. Important of the original focus of exploration has been on the software side of E-learning . Experimenters feel to agree that the use of tablets has implicit that's underutilized in academiam. The platoon reviewed publications of trials on the use of tablets in advanced education relating some crucial findings. From these findings, the platoon has developed system conditions for the design of a purpose- erected educational tablet for advanced education. These chancing include opting tablets that would image the

institution's own computer lab designs, including the operating system, educational operations, as well as confining use of camera, confining download of none-educational operations and apps. In addition, the platoon recommends the tablet to be pre-loaded with precisely named educational apps that would round and use the institution's own dispatch system in a form of case communication link to the tablet to enhance the benefit of the device as a dynamic and active literacy tool.

3.2.1 Non-Functional requirement

Requirement Analysis, also known as Requirement Engineering, is the process of defining user expectations for a new software being built or modified. In software engineering, it is sometimes referred to loosely by names such as requirements gathering or requirements capturing. Requirements analysis encompasses those tasks that go into determining the needs or conditions to meet for a new or altered product or project, taking account of the possibly conflicting requirements of the various stakeholders, analyzing, documenting, validating and managing software or system requirements. Types of Non-functional Demand :

- Usability demand.
- Utility demand.
- Manageability demand.
- Recover ability demand.
- Security Demand.
- Data Integrity demand.
- Capacity demand.
- Vacuity demand.

3.2.2 Functional requirement

A Functional Requirement (FR) is a description of the service that the software must offer. It describes a software system or its component. ... It can be a calculation, data manipulation, business process, user interaction, or any other specific functionality which defines what function a system is likely to perform.

Functional Requirements of our system :

- Sign Up.
- Sign In.
- View Course Information.
- Search Course Information.
- Enroll Course
- Download Book
- Read Books
- Watch Course Video and PPT
- Attend Quiz Test
- See Test Results
- Upload Course along with it's materials
- Upload Books
- Update Course Content

3.3 Methodology

3.3.1 Agile Development model:

We choose Agile Development model for our project. Agile Development model is a faster way to develop any software. It is people and result focused methodology. It's flexible and easy to improve the quality of software. It works on the principle of fast and incremental way.

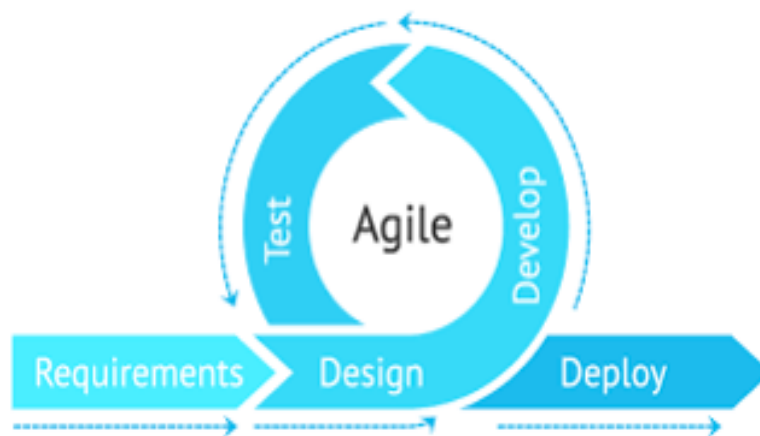


Figure 3.1: Agile Development model

Reasons for choosing Agile Development model:

- Clients are happier with the end product. This is possible due to improvements and involving clients in development decisions throughout the process;
- Lower cost; Fast releases;
- More open communication between the teams and clients;
- Greater project transparency. It is achieved due to regular meetings with the clients and systems. That allows all the involved parties to access the project data and progress;
- Competitive advantage to the team thanks to spotting defects and making changes throughout the development process rather than at the end;
- There are some high risk features and goals.

3.3.2 Gantt Chart

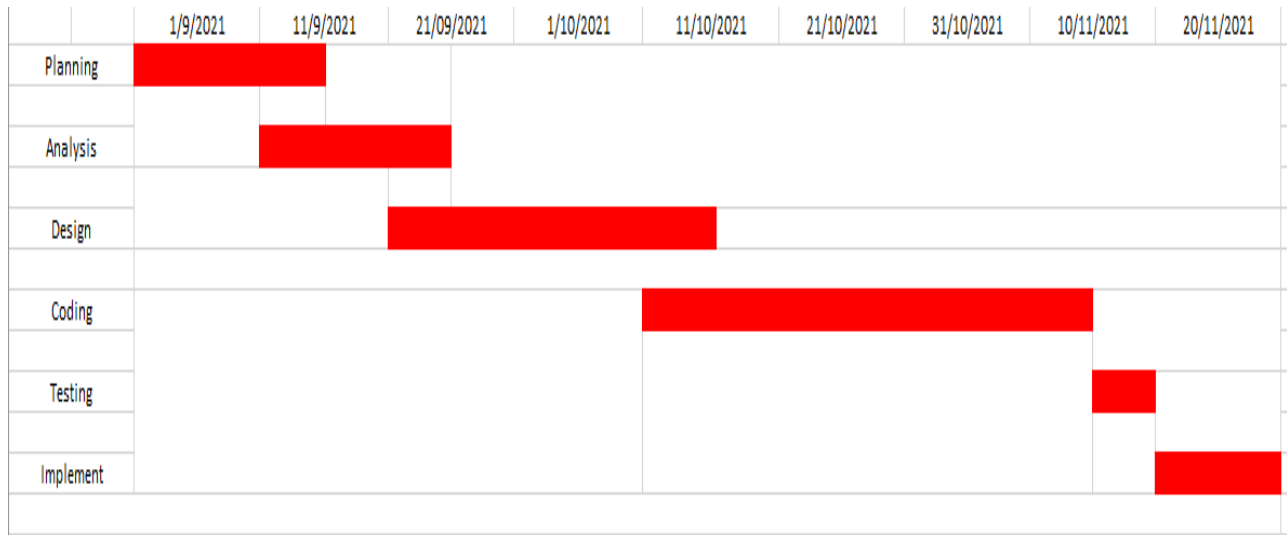


Figure 3.2: Context Level DFD of Lernen

3.3.3 Data Flow Diagram

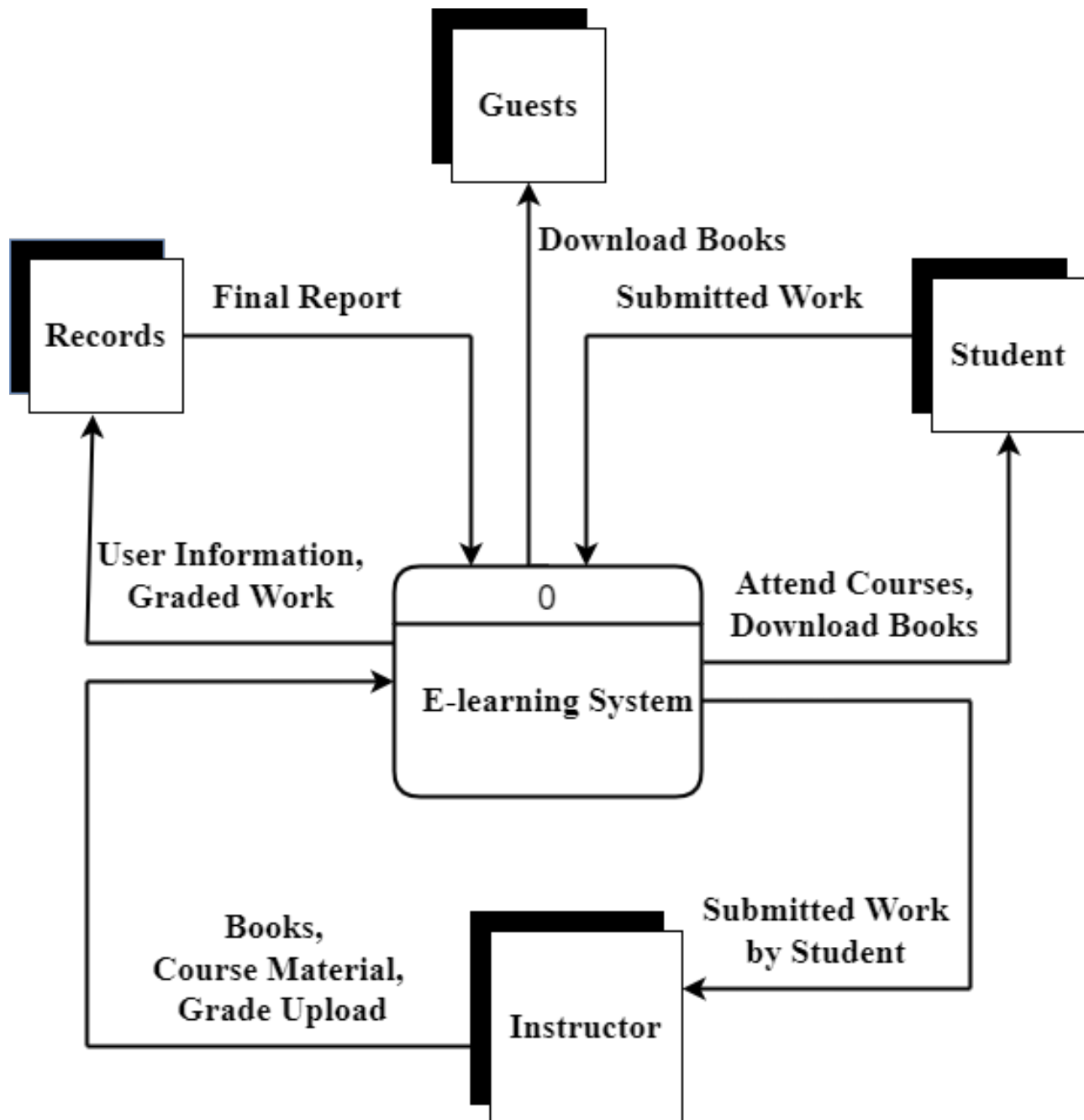


Figure 3.3: Context Level DFD of Lernen

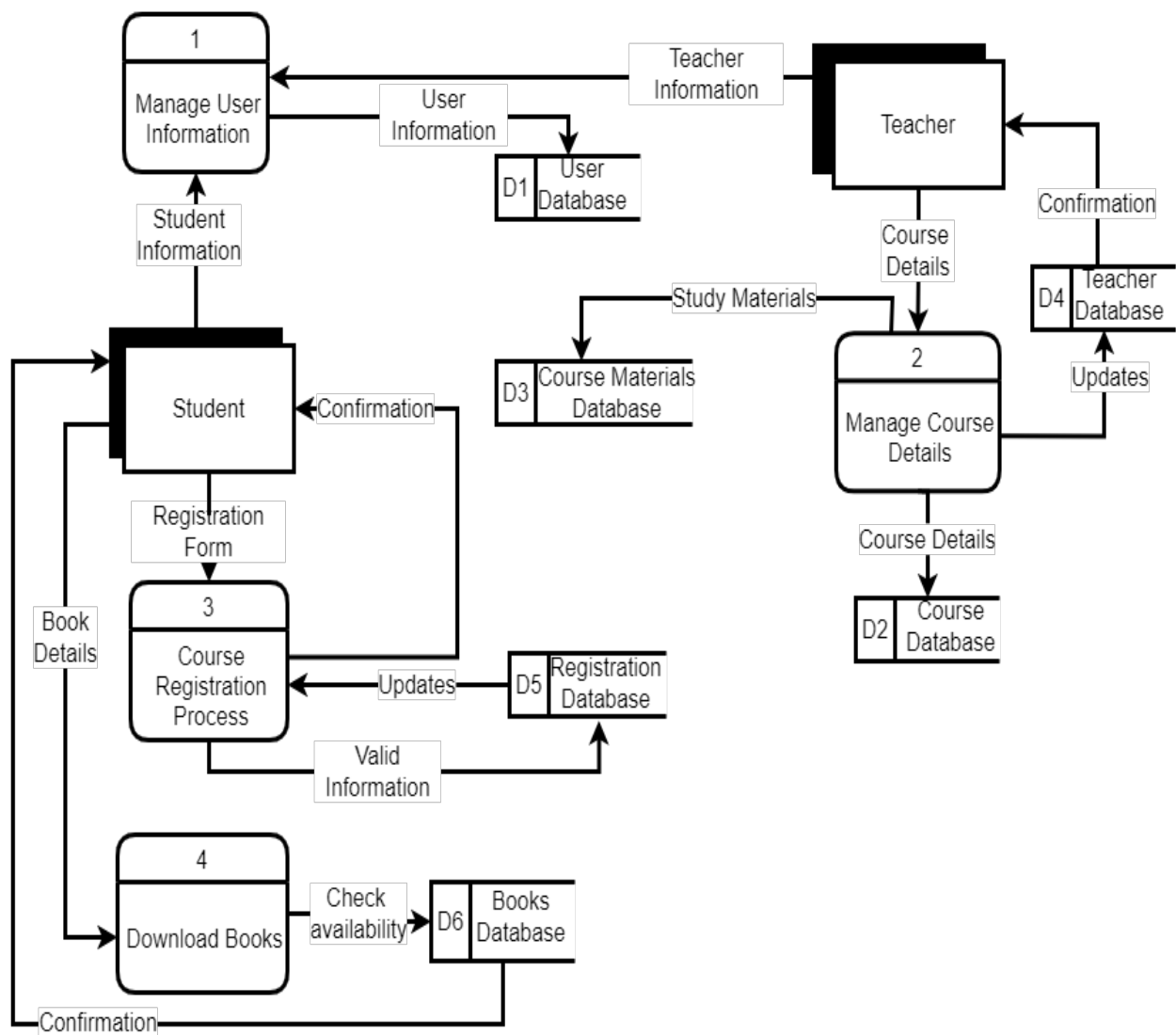


Figure 3.4: Level-0 DFD of Lernen

3.3.4 ER Diagram

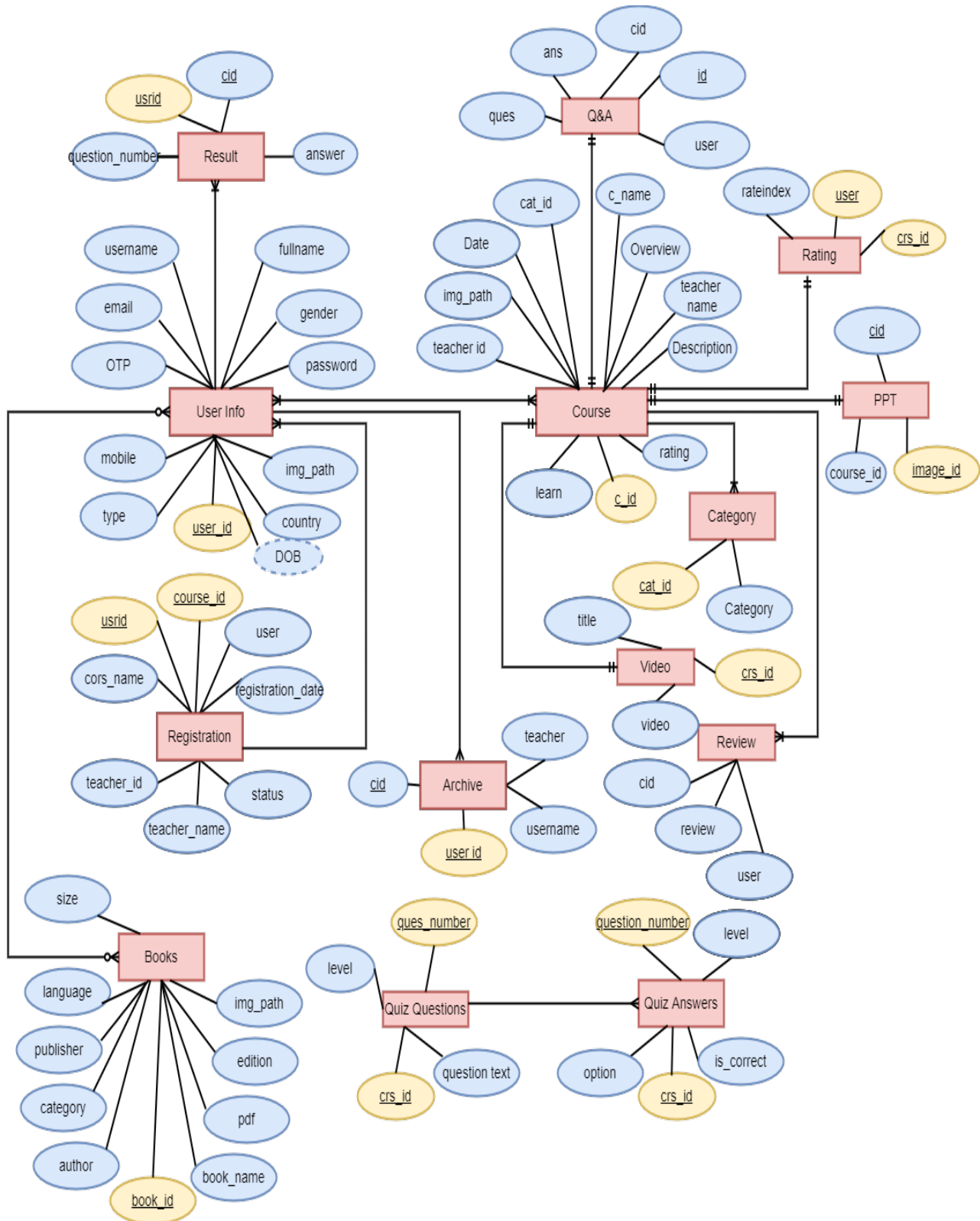


Figure 3.5: ER Diagram

3.3.5 Use Case Diagram

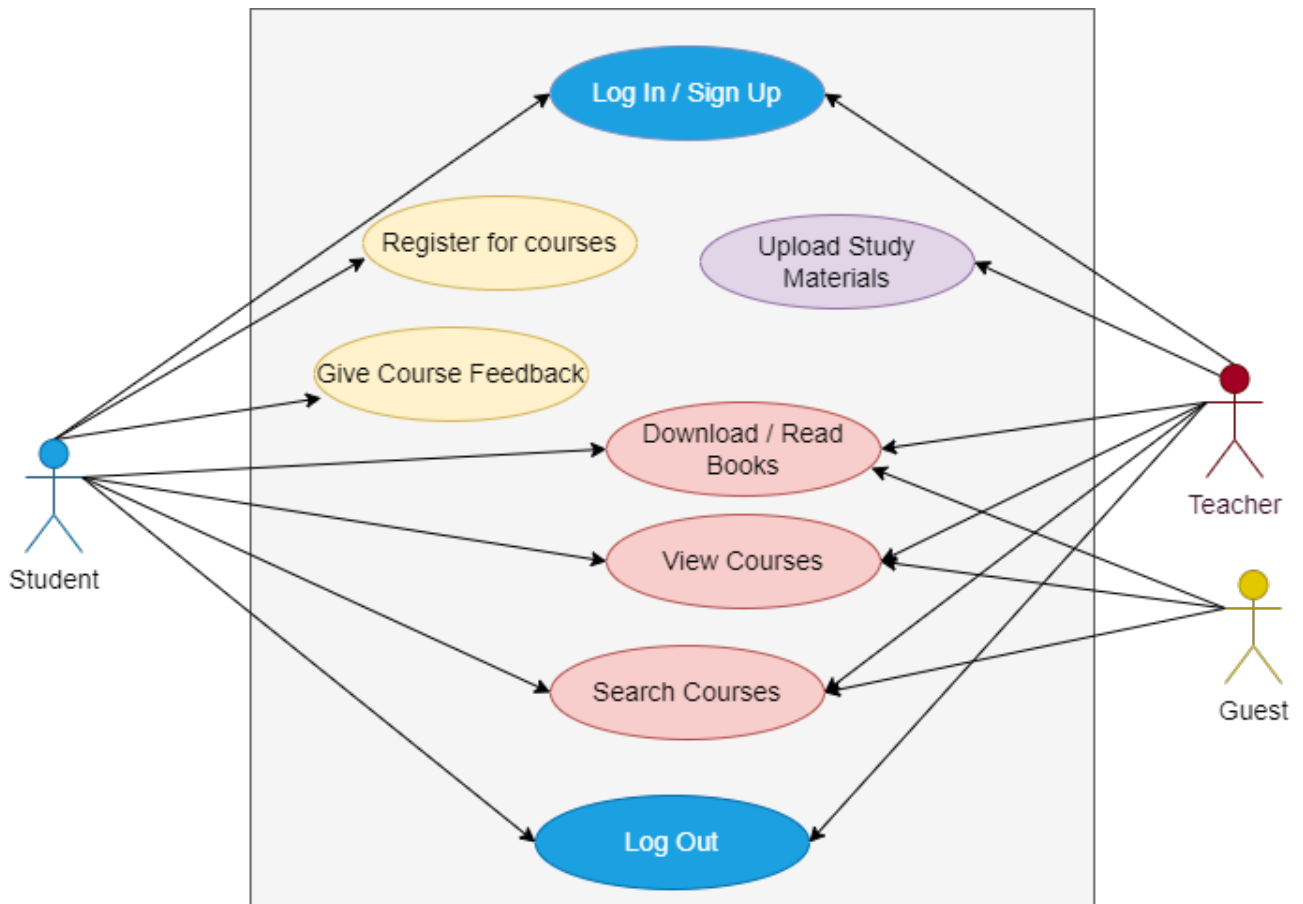


Figure 3.6: Use case Diagram

3.3.6 Activity Diagram

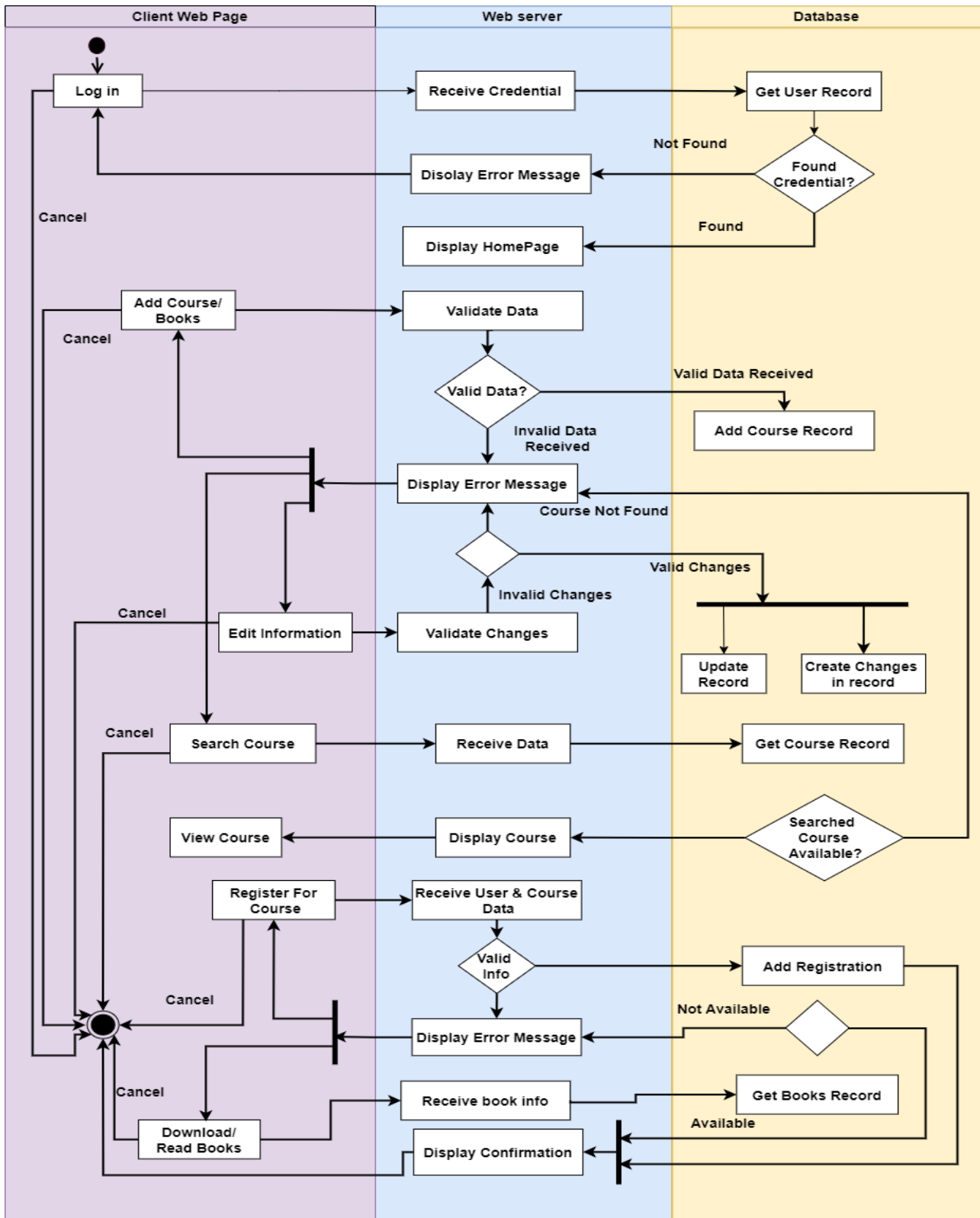


Figure 3.7: Activity Diagram

3.3.7 Sequence Diagram

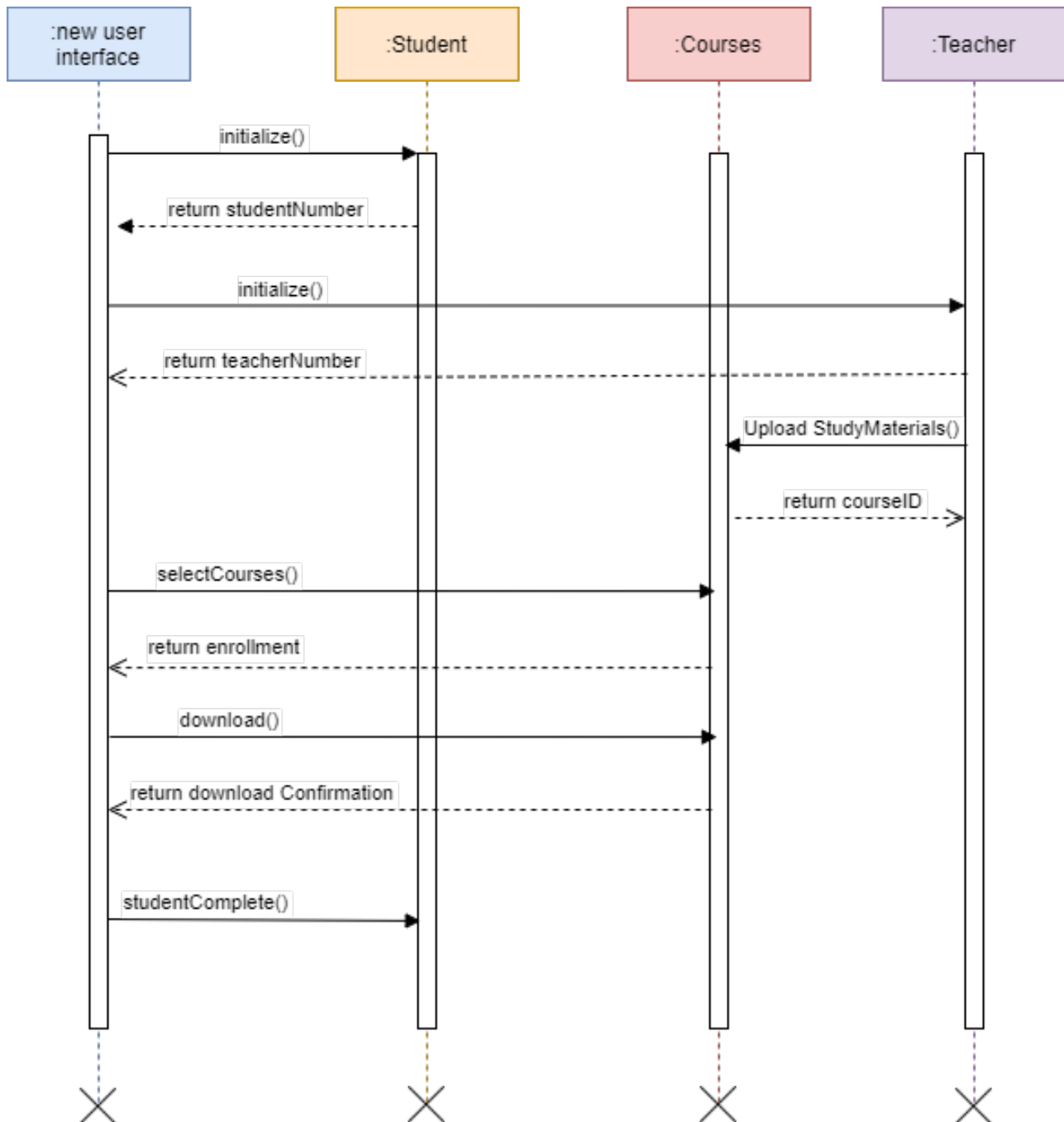


Figure 3.8: Sequence Diagram

3.3.8 Class Diagram

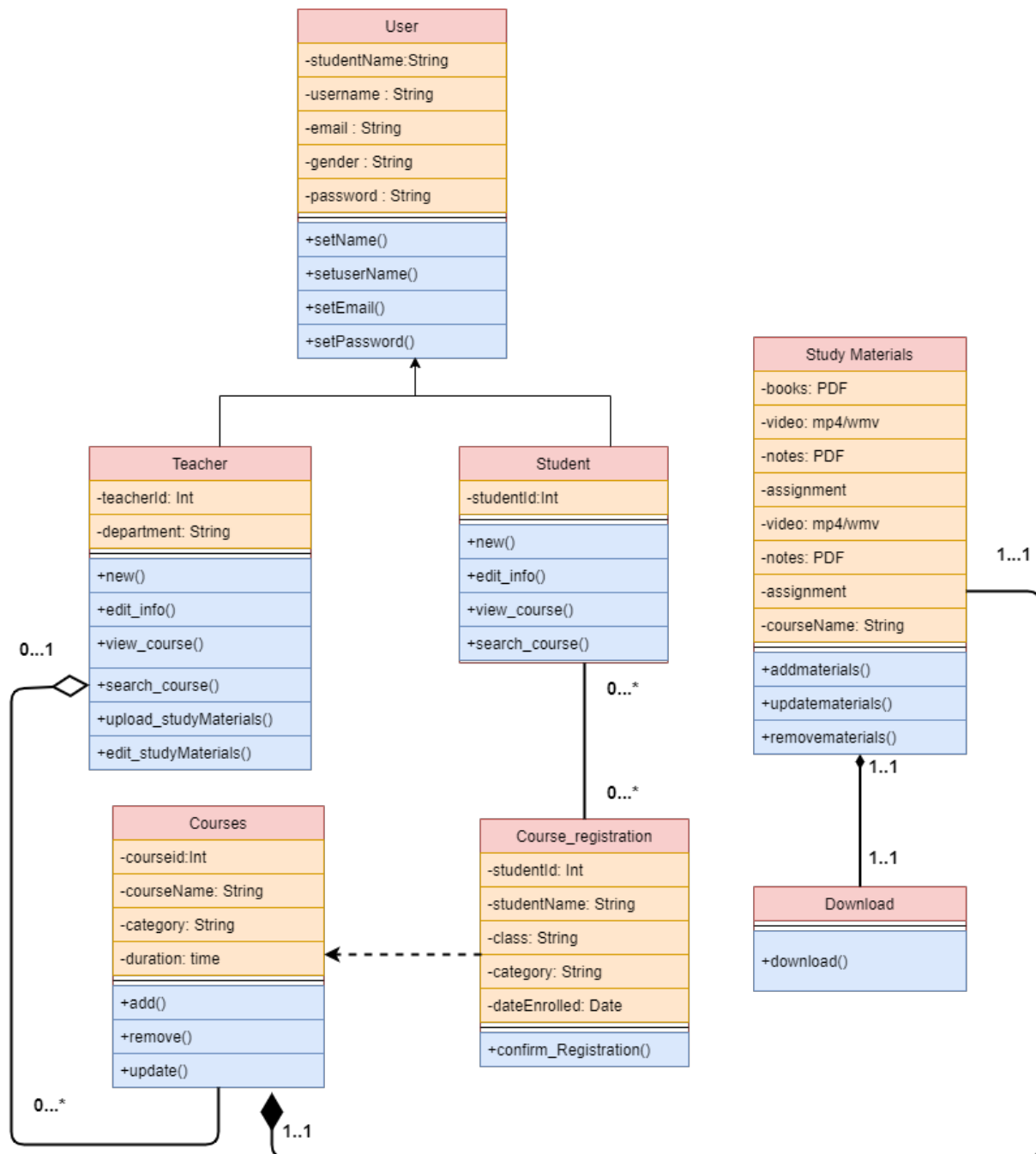


Figure 3.9: Class Diagram

3.3.9 Database Design

+ Options							
cors_name	c_id	register_date	tcr_id	tcr_name	user	usrid	status
PHP	1	2021-11-10	2	Humaira Ahmed	shirin	1	done
Graphic Design	10	2021-11-11	2	Humaira Ahmed	shirin	1	In progress
Web Design	4	2021-11-11	9	Shirin Sultana	shirin	1	In progress
Develop your own game	14	2021-11-11	2	Humaira Ahmed	shirin	1	In progress
PHP	1	2021-11-11	2	Humaira Ahmed	maruf	10	In progress
PHP	1	2021-11-11	2	Humaira Ahmed	ria	7	In progress
Frontend web development Course	15	2021-11-11	9	Shirin Sultana	maruf	10	In progress
Frontend web development Course	15	2021-11-12	9	Shirin Sultana	humaira	2	In progress
PHP	1	2021-11-12	2	Humaira Ahmed	shirinria	9	In progress
PHP	1	2021-11-15	2	Humaira Ahmed	humaira	2	done

Figure 3.10: Registration

category	c_id	c_name	Date	des	img_path	tid	tname	cdes	learn	rating
13	1	PHP	2021-11-03	PHP is a general-purpose scripting language geared...	uploads/PHP.jpg	2	Humaira Ahmed	Are you new to PHP or need a refresher? Then this ...	You will learn to create a (CMS) Content Managemen...	3
12	2	IELTS	2021-11-03	The International English Language Testing System....	uploads/IELTS.png	9	Shirin Sultana		NULL	1
2	4	Web Design	2021-11-03	Web design encompasses many different skills and d...	uploads/Web Design.jpg	9	Shirin Sultana		NULL	0
2	10	Graphic Design	2021-11-09	Graphic design is the profession and academic disc...	uploads/Graphic Design.jpg	2	Humaira Ahmed		NULL	2
1	14	Develop your own game	2021-11-11	Sometimes game development is a challenging, painf...	uploads/Develop your own game.jpg	2	Humaira Ahmed	Game Development is the art of creating games and ...	Master beginner C# concepts, like variables, "if" ...	0
1	15	Frontend web development Course	2021-11-11	We'll Cover HTML5, CSS3, JavaScript, jQuery, Boots...	uploads/Frontend web development Course.jpg	9	Shirin Sultana	Web development is the work involved in developing...	You'll learn all the fundamentals of Front-End Web...	3.33333333333333

Figure 3.11: Courses

aname	bcat	bid	bname	pdf	publshr	edtn	size	img_path	lang
Weygandt Kimmel Lieso	Academic(College)	9	Accounting	Accounting Principles, 10th edition.pdf	John Wiley & Sons, Inc	10th edition	28188938	uploads/download (2).jpg	English
Vishal Layka	Development	10	Learn Java for Web Development	Learn Java for Web Development .pdf			NULL	uploads/java.jpg	English
Noel Kalicharan	Programming	14	Learn to Program with C	Learn to Program with C_ Learn to Program using th...			3	uploads/c.jpg	English
V.V.K. Subburaj	GK	15	Basic General Knowledge Book	Basic General Knowledge Book (PDFDrive).pdf	Sura College of Competition	(1 January 2012)	5	uploads/Basic General Knowledge Book.jpg	English
Monira Khaton	IELTS	16	Best Practice Book for IELTS Writing 230 IELTS Wri...	Best Practice Book for IELTS Writing 230 IELTS Wri...			4	uploads/41i809t4r+L.jpg	
Glenn Beck	Islam Shikkha	17	It IS About Islam	It IS About Islam_ Exposing the Truth About ISIS, ...		August 18, 2015	2	uploads/It IS About Islam_ Exposing the Truth Abou...	English
Rupayon Neogy	Design	18	Web Design & UI_UX	Web Design & UI_UX (PDFDrive).pdf			3	uploads/html.jpg	English
Andy Ilachinski	Photography	19	BWClassNotesBelnavisGallery	BWClassNotesBelnavisGalleryFeb2011.pdf		Feb2011	6	uploads/1- 03c8a2bd48.jpg	English
Trishna	Academic(School)	20	Chemistry (Class 9)	Chemistry (Class 9) (PDFDrive).pdf		2014	33	uploads/382f5ec6691110	English

Figure 3.12: Books

Chapter 4

DESIGN IMPLEMENTATION

4.1 User Section

We have three kinds of users,

- Guest
- Learner
- Instructor

We will show the activities of all kind of users in this part.

4.1.1 Database Connection

```
<?php
$host='localhost'; $db='learnen_two'; $user='root'; $pass=''; $charset='utf8mb4';
$dsn = "mysql:host=$host;dbname=$db;charset=$charset";
try {
    $pdo = new PDO($dsn , $user , $pass);
    $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
}
catch (PDOException $e) {
    throw new PDOException($e->getMessage());
}
?>
```

Figure 4.1: Database Connection

4.1.2 Guest Home Page



Figure 4.2: Guest Homepage

4.1.3 Registered user's Home Page

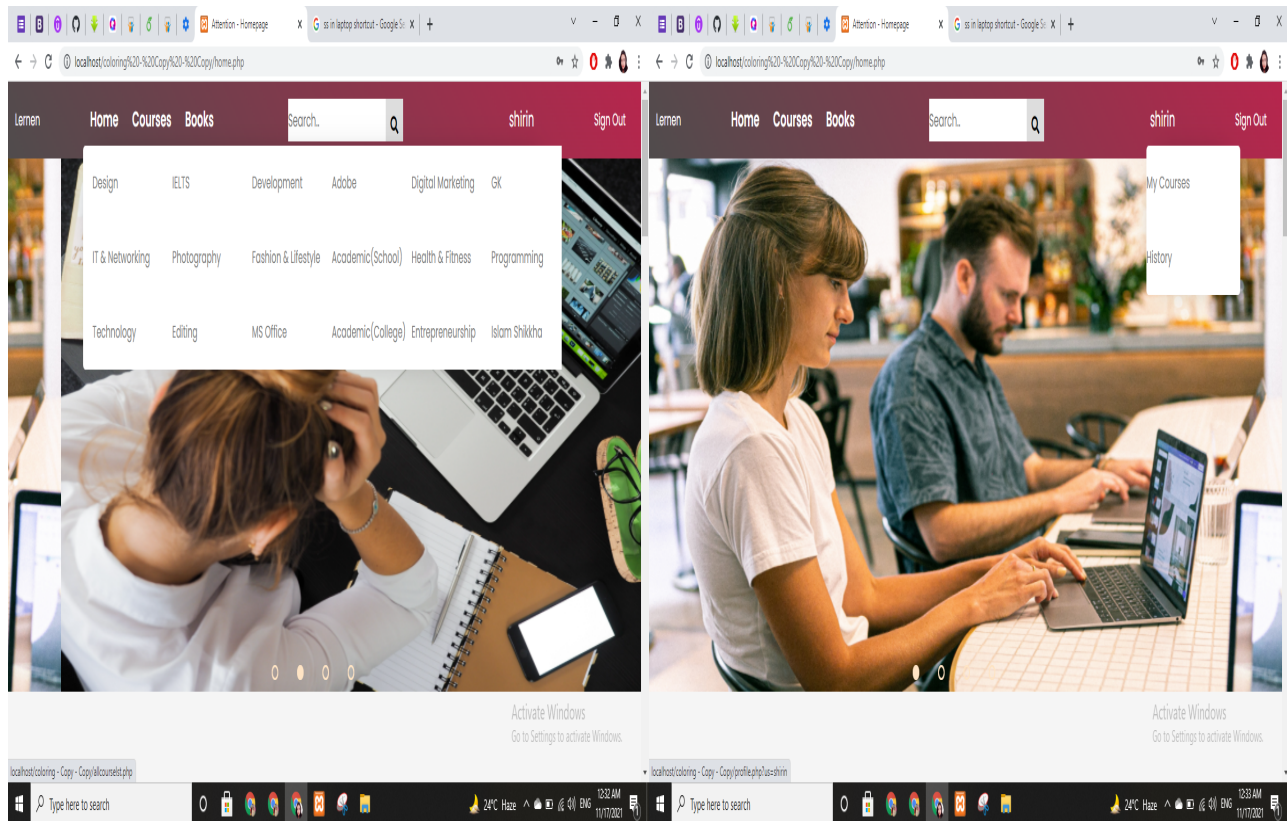


Figure 4.3: Registered user's homepage

4.1.4 Home Page Backend

```
public function num(){
    try {
        $sql = "select * from sign ";
        $stmt = $this->db->prepare($sql);
        $stmt->execute();
        $result = $stmt->rowCount();
        echo $result;
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function tnum(){
    try {
        $sql = "select * from sign where type= 'Teacher' ";
        $stmt = $this->db->prepare($sql);
        $stmt->execute();
        $result = $stmt->rowCount();
        echo $result;
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function lnum(){
    try {
        $sql = "select * from sign where type= 'Learner' ";
        $stmt = $this->db->prepare($sql);
        $stmt->execute();
        $result = $stmt->rowCount();
        echo $result;
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function cnum(){
    try {
        $sql = "select * from course ";
        $stmt = $this->db->prepare($sql);
        $stmt->execute();
        $result = $stmt->rowCount();
        echo $result;
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.4: Home Page Backend

```
public function crd(){
    try{
        $sql = "select * from course where month(Date)=month(now());";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.5: Home Page Backend

4.1.5 User Login

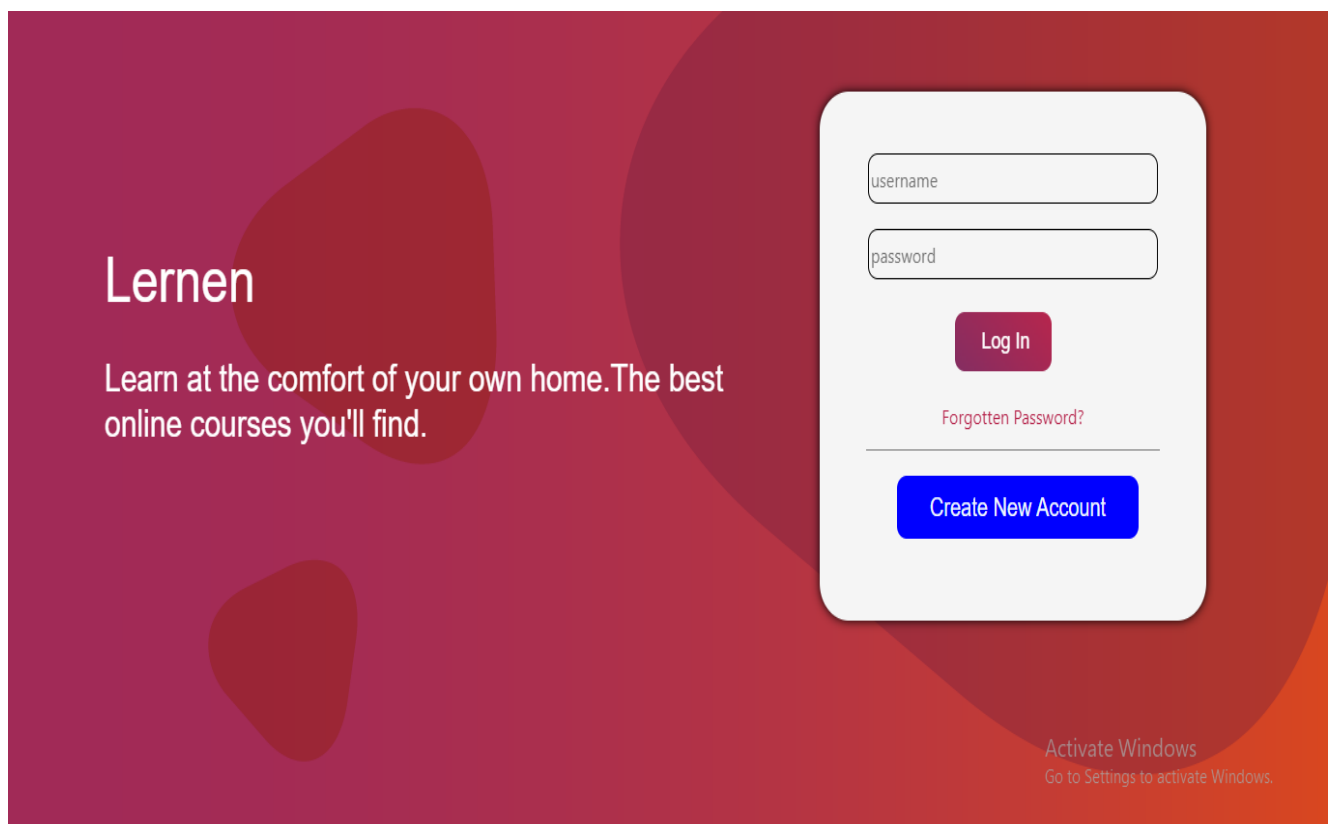


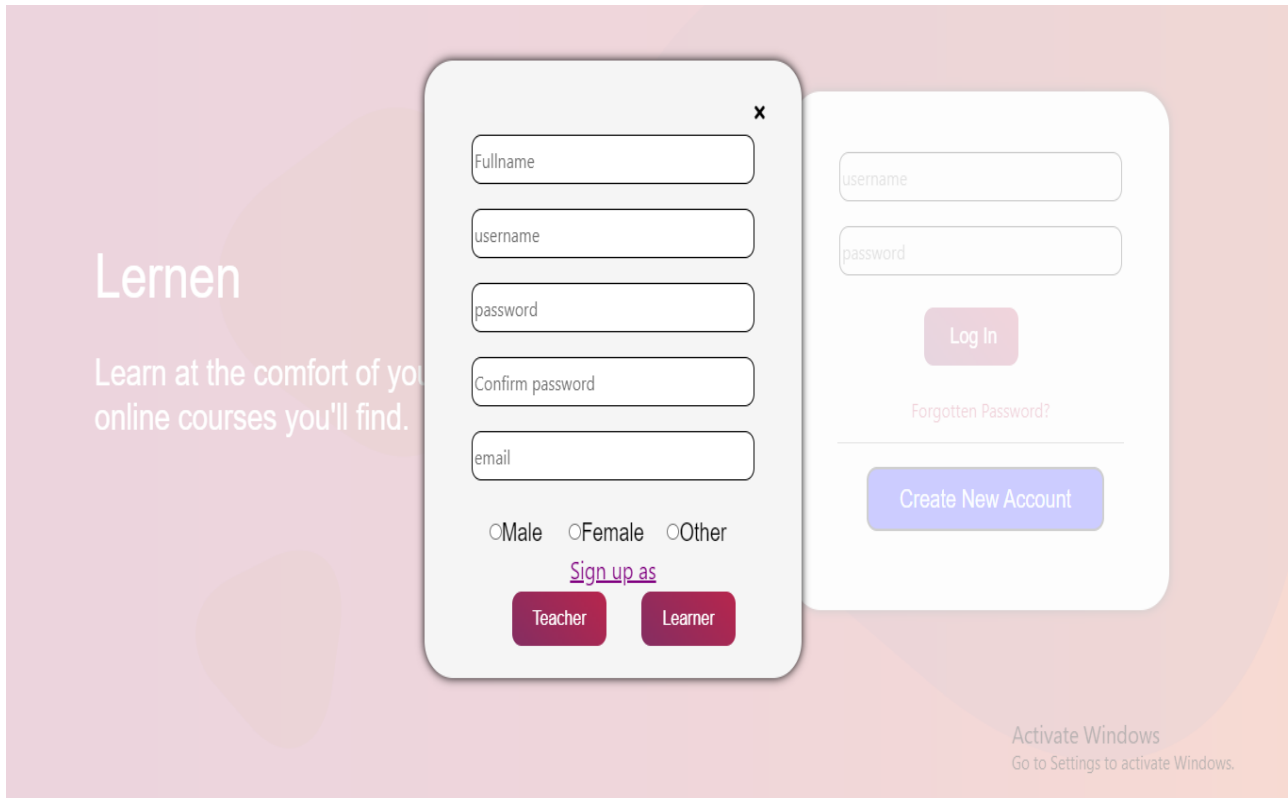
Figure 4.6: users Login

4.1.6 User Login Backend

```
public function getuser($user,$pass){
    try {
        $sql = "select * from sign where username = :user AND passwor = :pass";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':user',$user);
        $stmt->bindParam(':pass',$pass);
        $stmt->execute();
        $result=$stmt->fetch();
        return $result;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
public function getem($em){
    try {
        $sql = "select * from sign where username = :user AND otpmtch='Not Match'";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':user',$em);
        $stmt->execute();
        $result=$stmt->fetch();
        return $result;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.7: users Login Backend

4.1.7 Sign Up



The image shows a user registration interface. On the left, a light purple sidebar contains the text "Lernen" and "Learn at the comfort of your own home. Find the online courses you'll find." Below this is a registration form with a close button (X) in the top right corner. The form includes input fields for "Fullname", "username", "password", "Confirm password", and "email". Below these fields are three radio buttons labeled "Male", "Female", and "Other". A link "Sign up as" is positioned above two red buttons labeled "Teacher" and "Learner". To the right of the sidebar is a white registration card. It contains input fields for "username" and "password", a "Log In" button, a "Forgotten Password?" link, and a "Create New Account" button. At the bottom right of the page, there is a "Activate Windows" watermark with the text "Go to Settings to activate Windows."

Lernen

Learn at the comfort of your own home. Find the online courses you'll find.

Fullname

username

password

Confirm password

email

☐ Male ☐ Female ☐ Other

[Sign up as](#)

Teacher Learner

username

password

Log In

[Forgotten Password?](#)

Create New Account

Activate Windows
Go to Settings to activate Windows.

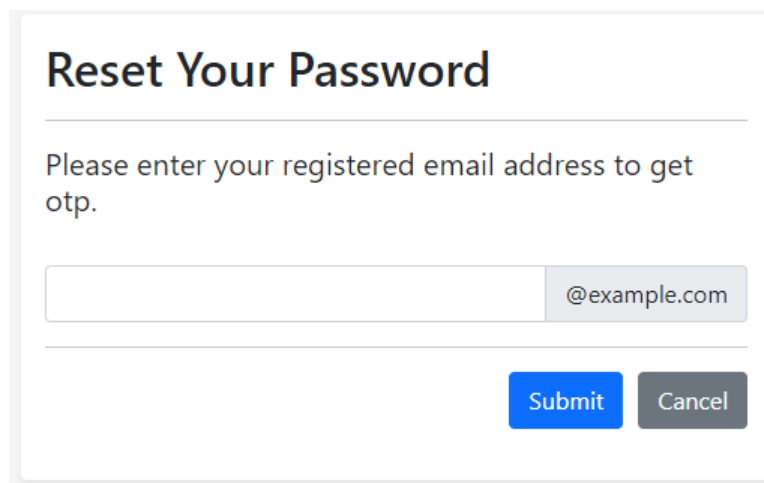
Figure 4.8: New users registration

4.1.8 Sign Up Backend

```
public function insert($email,$otp,$user){
    try {
        $sql = "select * from sign where email = :email";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':email',$email);
        $stmt->execute();
        $reslt = $stmt->rowCount();
        if($reslt==0)
        {
            $sql = "INSERT INTO sign (email,OTP,username) VALUES (:email, :otp, :user)";
            $stmt = $this->db->prepare($sql);
            $stmt->bindParam(':email',$email);
            $stmt->bindParam(':otp',$otp);
            $stmt->bindParam(':user',$user);
            $stmt->execute();
            return true;
        }
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

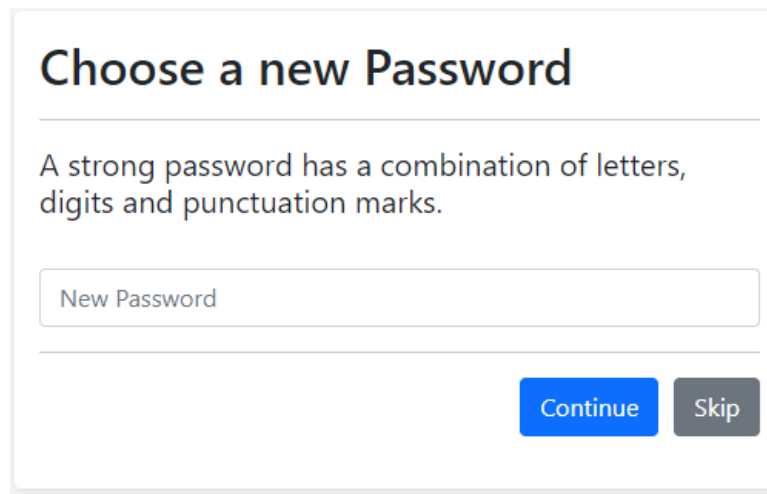
Figure 4.9: Sign Up Backend

4.1.9 Forget Password



The image shows a web form titled "Reset Your Password". Below the title is a horizontal line. The text "Please enter your registered email address to get otp." is displayed. There is a text input field with a placeholder "@example.com". Below the input field is another horizontal line. At the bottom right, there are two buttons: a blue "Submit" button and a grey "Cancel" button.

Figure 4.10: Enter registered email to get OTP



Choose a new Password

A strong password has a combination of letters, digits and punctuation marks.

Figure 4.11: Set new password

4.1.10 Forget Password Backend

```

<?php
include_once 'includes/session.php';
require_once 'db/conn.php';
require_once 'sendmail.php';
if(isset($_POST['forgotpass'])){
    $email= $_POST['email'];
    $sub="VERIFICATION";
    $otp=rand(1000,99999);
    $res=$login->getemail($email,$otp);
    if($res){
        sendmail ::sendmail($email,$full,$sub,$otp);
        $usr = $res['username'];
        $_SESSION['user'] = $usr;
        $_SESSION['email'] = $email;
        header("Location: forgotsuccess.php?usr=$usr&eml=$email");
    }
    else
        echo '<div class ="alert alert-danger" id= "alert">This email is not registered</div>';
}
if(isset($_POST['cncl']))
    header("Location: index1.php");
?>

```

```

public function getemail($email,$otp){
    try {
        $sql = "select * from sign where email = :email";
        $stmt = $this->db->prepare($sql);

        $stmt->bindParam(':email',$email);
        $stmt->execute();
        $result=$stmt->fetch();
        $reslt = $stmt->rowCount();
        if($reslt>0){
            $sql2 = "UPDATE `sign` SET `otp`=:otp WHERE email=:email";
            $stmt = $this->db->prepare($sql2);
            $stmt->bindParam(':email',$email);
            $stmt->bindParam(':otp',$otp);
            $stmt->execute();
        }
        return $result;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

```

Figure 4.12: Get Email Backend

```

<?php
$title = 'Index';
require_once 'includes/header.php';
require_once 'db/conn.php';
require_once 'sendmail.php';
$usser = $_SESSION['user'];
$email = $_SESSION['email'];
if (isset($_POST['rsnd'])) {
    $vuser = $_POST['ouser'];
    $email = $_POST['remail'];
    $full = $_POST['rfull'];
    $newotp = rand(1000, 99999);
    $res = $opt->u_otp($vuser, $newotp);
    $sub = "VERIFICATION";
    if ($res) {
        sendmail :: sendmail($email, $full, $sub, $newotp);
    } else {
        echo 'failupdt';
    }
}
if (isset($_POST['sotp'])) {
    $vuser = $_POST['ouser'];
    $o = $_POST['otp'];
    $result = $opt->otp($email, $o);
    if ($result) {
        $user = $_SESSION['user'];
        header("Location: rpass.php");
    }
    else {
        echo '<div class="alert alert-danger" id="alert">OTP doesnot match</div>';
    }
}
?>

```

```

public function u_otp($email, $otp) {
    try {
        $sql = "UPDATE `sign` SET `OTP`=:otp WHERE email=:email ";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':email', $email);
        $stmt->bindParam(':otp', $otp);
        $stmt->execute();
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

```

Figure 4.13: Get otp Backend

```

<?php
$title = 'Index';
include_once 'includes/session.php';
require_once 'db/conn.php';
$usser = $_SESSION['user'];

if (isset($_POST['submit'])) {
    $pass = $_POST['newpass'];
    $newpass = md5($pass);
    $success = $login->updtpass($usser, $newpass);
    if ($success) {
        $_SESSION['user'] = $user;
        $_SESSION['full'] = $full;
        $_SESSION['email'] = $email;
        $_SESSION['tp'] = $type;
        $_SESSION['gn'] = $gender;
        $_SESSION['pass'] = $newpass;
        header("Location: index1.php");
    }
}
if (isset($_POST['cnc1'])) {
    header("Location: index1.php");
}
?>

```

```

public function updtpass($usser, $newpass) {
    try {
        $sql = "UPDATE `sign` SET `passwor`=:newpass WHERE username=:usr";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':usr', $usser);
        $stmt->bindParam(':newpass', $newpass);
        $stmt->execute();
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

```

Figure 4.14: Reset Password Backend

4.1.11 Course List Page

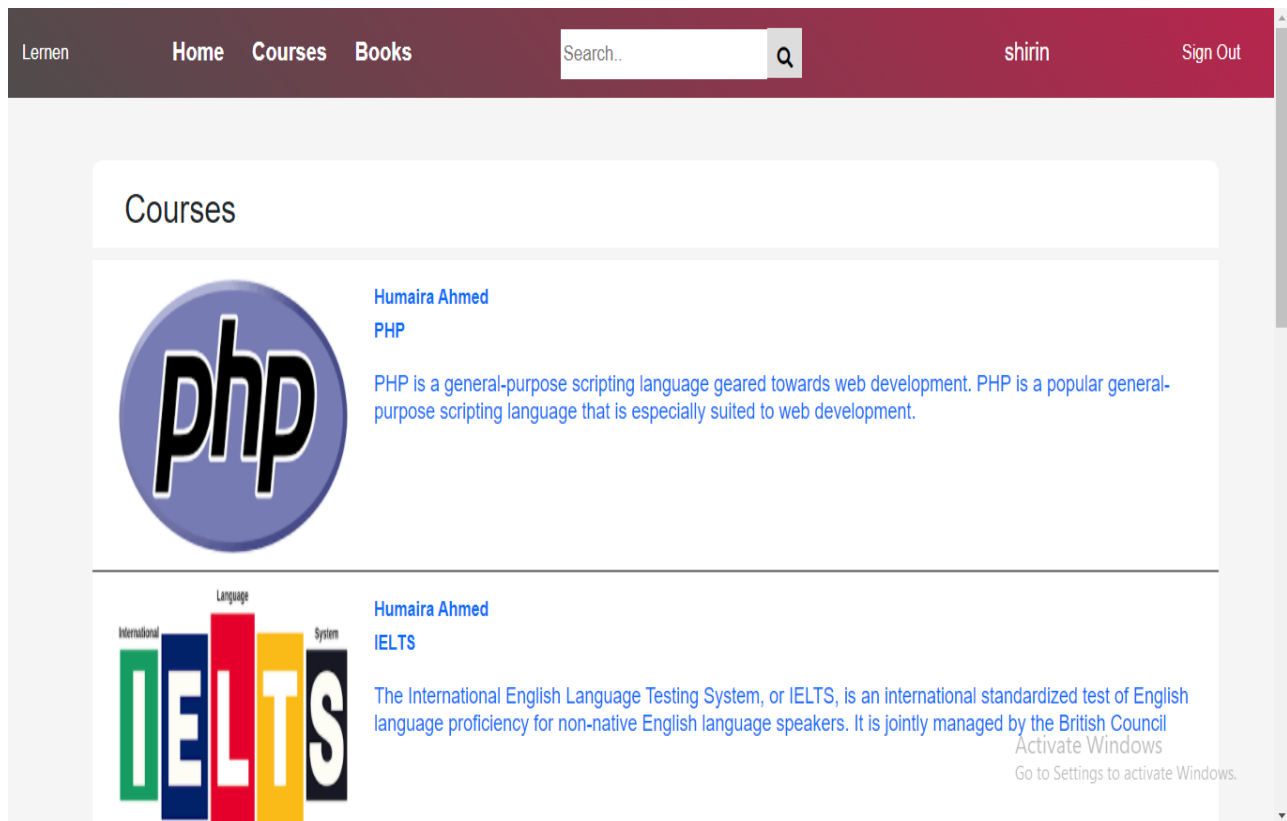


Figure 4.15: Course List

4.1.12 Course Page Backend

```
public function allcrslst(){
    try{
        $sql = "SELECT * FROM `course` ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function crslst($cat){
    try{
        $sql = "SELECT * FROM `course` where category=$cat";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.16: Course Page Backend

4.1.13 Search Course

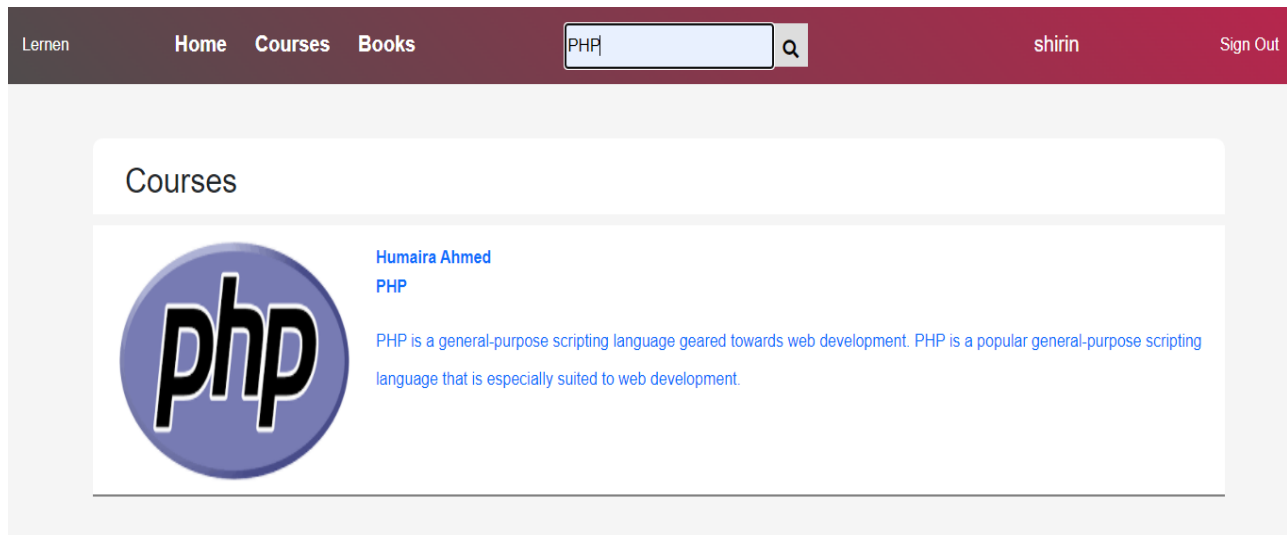


Figure 4.17: Search Course

4.1.14 Search Course Backend

```
if(isset($_POST['search']))  
{  
    $host='localhost';  
    $db='learnen_two';  
    $user='root';  
    $pass='';  
    $charset='utf8mb4';  
  
    $conn =new mysqli ( "localhost","root","", "learnen_two");  
    $data= $_POST['serch'];  
    $sql = "select * from course where match (c_name, des) against ('$data')";  
    // $sql = "SELECT * FROM `course` WHERE c_name=:data";  
    $result = mysqli_query($conn, $sql);  
    // $executerrecord=$result->execute(array(":data"=>$data));  
    echo '<div class="crslist"><div class="lsthead">  
<h2><span>&nbsp;</span>Courses</h2>  
</div></div>';  
    while($row = mysqli_fetch_assoc($result)){  
        $title = $row['c_name'];  
        $desc = $row['des'];  
        $tcr = $row['tname'];  
        $thread_id= $row['c_id'];  
        $imgg=$row['img_path'];  
    }  
}
```

Figure 4.18: Search Course Backend

4.1.15 Books

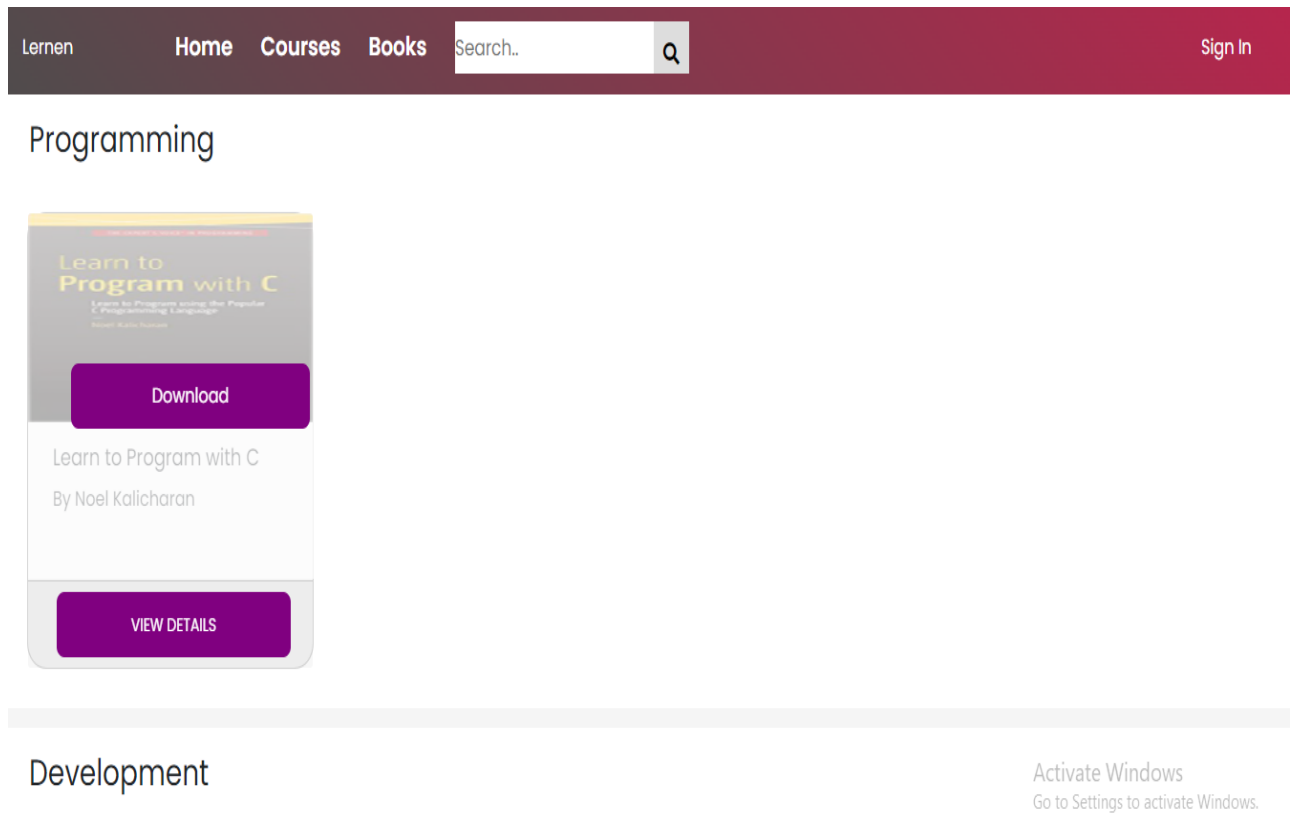


Figure 4.19: Books Page

4.1.16 Books Backend

```
public function best(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='Programming' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
public function devlpmnt(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='Development' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
public function genknw(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='GK' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
public function ilts(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='IELTS' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
public function islamic(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='Islam Shikkha' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
public function design(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='Design' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.20: Books Page Backend

```
public function school(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='Academic(School)' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function college(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='Academic(College)'";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function english(){
    try{
        $sql = "SELECT * FROM `books`WHERE bcat='Photography' ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.21: Books Page Backend

4.1.17 Book Details

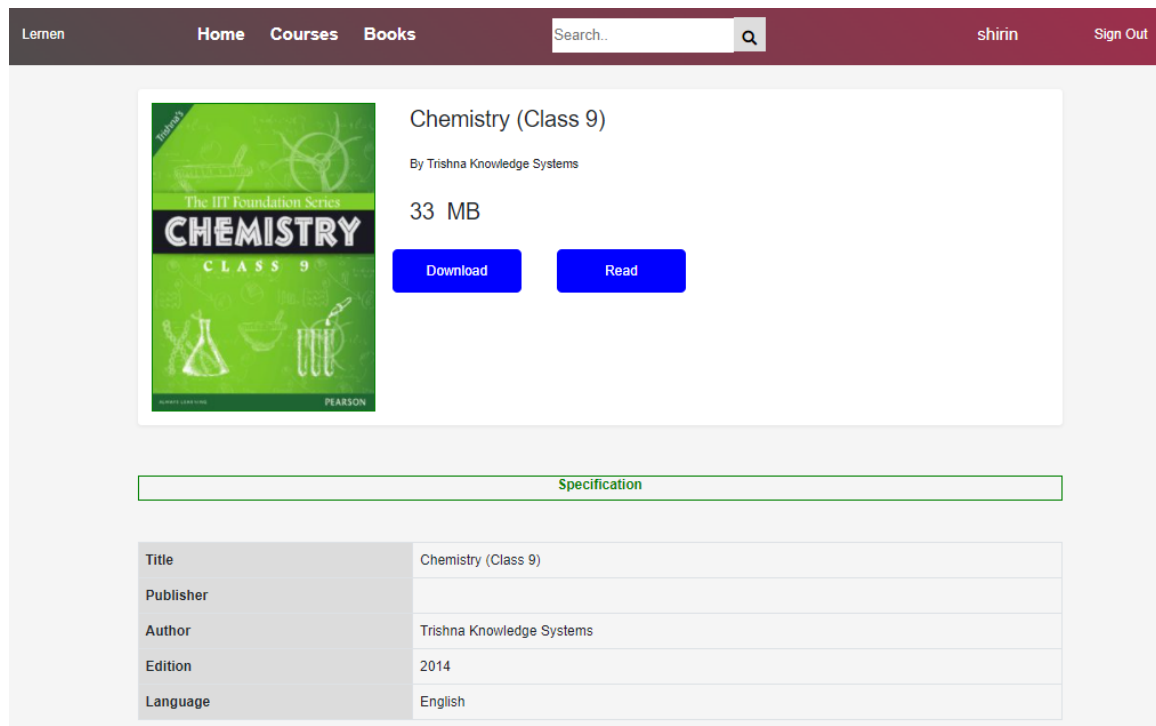


Figure 4.22: Books Details Page

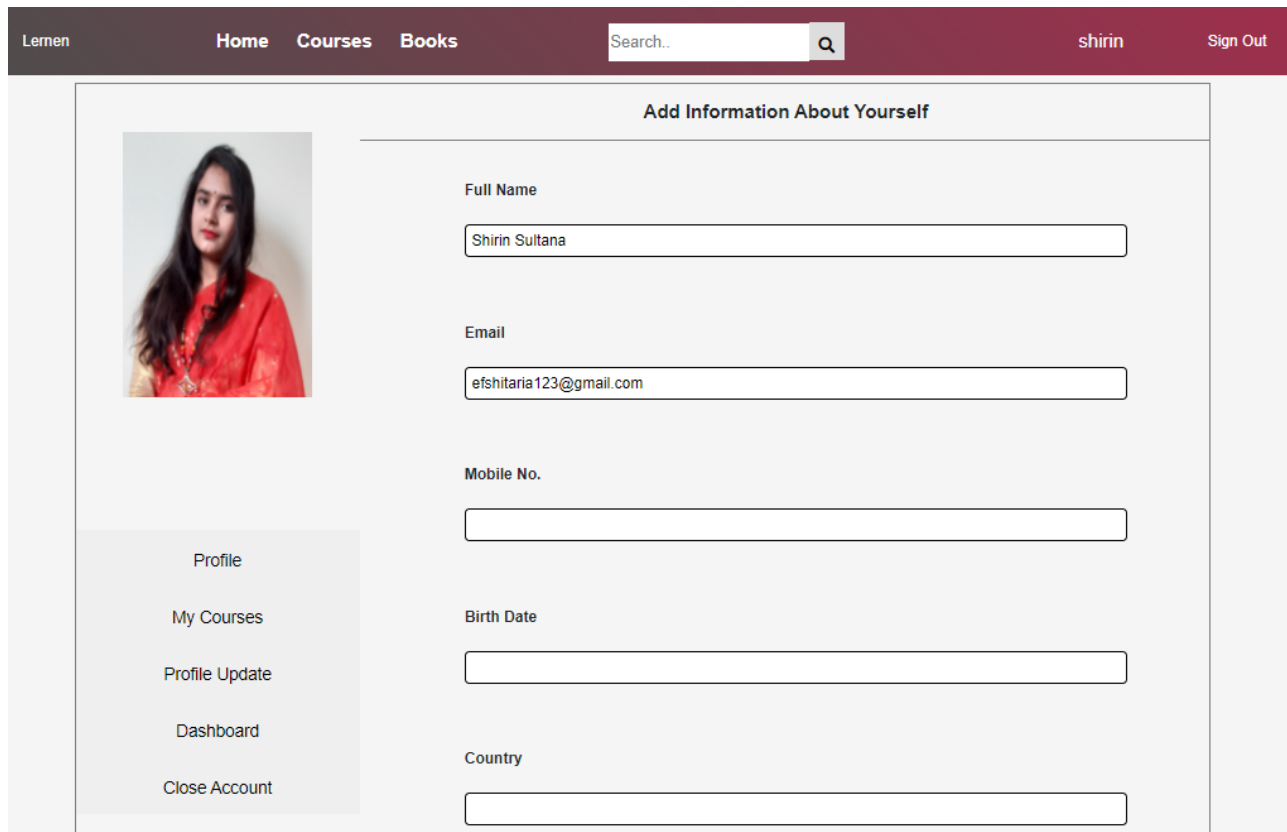
4.1.18 Book Details Backend

```
public function bkdata($id){
    try{
        $sql = "select * from books where bid = :bid";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':bid', $id);
        $stmt->execute();
        $result = $stmt->fetch();
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.23: Books Details Backend

4.1.19 User Profile

Profile Info



The screenshot displays a web application interface for user profile management. At the top, a dark red navigation bar contains the text 'Lernen', 'Home', 'Courses', and 'Books'. A search bar with the placeholder 'Search..' and a magnifying glass icon is positioned to the right of these links. Further right, the username 'shirin' and a 'Sign Out' link are visible. Below the navigation bar, the main content area is divided into two sections. On the left, a vertical sidebar lists navigation options: 'Profile', 'My Courses', 'Profile Update', 'Dashboard', and 'Close Account'. The 'Profile' option is currently selected. The right section, titled 'Add Information About Yourself', contains a form with five input fields: 'Full Name' (pre-filled with 'Shirin Sultana'), 'Email' (pre-filled with 'efshitaria123@gmail.com'), 'Mobile No.', 'Birth Date', and 'Country'. A profile picture of a woman with long dark hair wearing a red garment is displayed to the left of the form fields.

Figure 4.24: New users registration

Dashboard

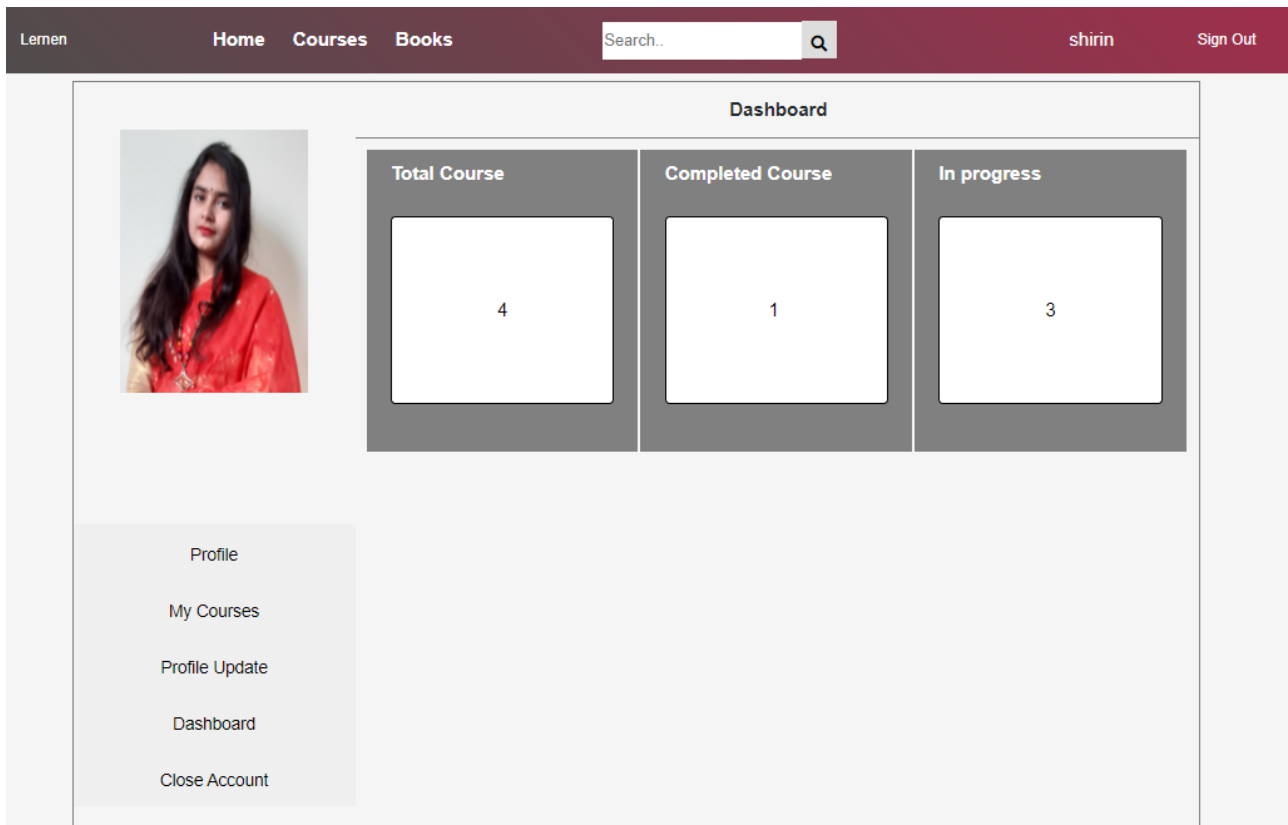


Figure 4.25: New users Dashboard

Close Account

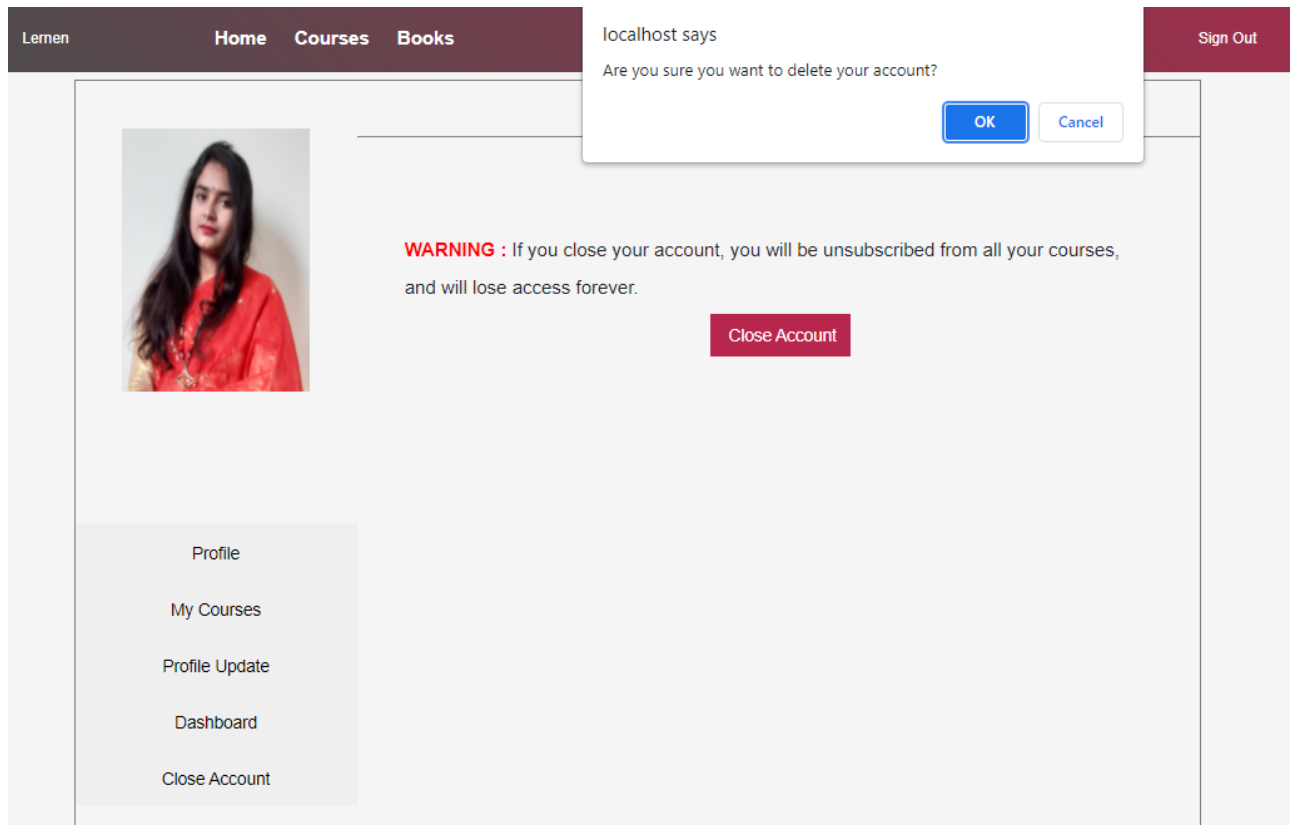


Figure 4.26: New User Close Account

4.1.20 User Profile Backend

```

public function data($user){
    try{
        $sql = "select * from sign where username = :user";
        $stmt = $this->db->prepare($sql); $stmt->bindParam(':user', $user);
        $stmt->execute();
        $result = $stmt->fetch();
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function delete($usr){
    try{
        $sql = "delete from sign where username = :user";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':user', $usr);
        $stmt->execute();
        return true;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function updtusrdata($n,$em,$mb,$bd,$cn,$destination,$ides){
    try{
        $sql = "UPDATE `sign` SET `mobile`=:mbl, `fullname`=:nam, `country`=:cntry, `bdate`=:dt,
        `img_path`=:img, `ides`=:ides WHERE email=:em";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':nam',$n); $stmt->bindParam(':mbl',$mb);
        $stmt->bindParam(':dt',$bd); $stmt->bindParam(':cntry',$cn);
        $stmt->bindParam(':em',$em); $stmt->bindParam(':ides',$ides);
        $stmt->bindParam(':img',$destination);
        $stmt->execute();
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

```

Figure 4.27: User Profile Backend

4.1.21 Enrolled Course

The screenshot displays the 'My Courses' interface. At the top, there is a red header bar with the text 'My Courses'. Below this, there are two tabs: 'All Courses' and 'Archived', with 'Archived' being the active tab. The main content area shows four course cards, each with a thumbnail image, a title, a creator name, and an 'Archive' link.

- Course 1:** Thumbnail shows the 'php' logo. Title: [PHP](#). Created by: [Humaira Ahmed](#). Link: [Archive](#).
- Course 2:** Thumbnail shows a man working on a laptop with the text 'LEARN GRAPHIC DESIGN AT HOME'. Title: [Graphic Design](#). Created by: [Humaira Ahmed](#). Link: [Archive](#).
- Course 3:** Thumbnail shows a laptop with the text 'WEB DESIGN'. Title: [Web Design](#). Created by: [Shirin Sultana](#). Link: [Archive](#).
- Course 4:** Thumbnail shows a yellow background with the text 'SO YOU WANNA BE A GAME DEVELOPER?'. Title: [Develop your own game](#). Created by: [Humaira Ahmed](#). Link: [Archive](#).

At the bottom right of the page, there is a Windows activation notice: 'Activate Windows. Go to Settings to activate Windows.'

Figure 4.28: My Course

4.1.22 Enrolled Course Backend

```
public function mycrs($uid){
    try{
        $sql = "SELECT * FROM registration,course where usrid=$uid AND registration.c_id=course.c_id";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function myarchv($uid){
    try{
        $sql = "SELECT * FROM archive,course where usr_id=$uid AND cid=c_id";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.29: Enrolled Course Backend

4.1.23 Course Content

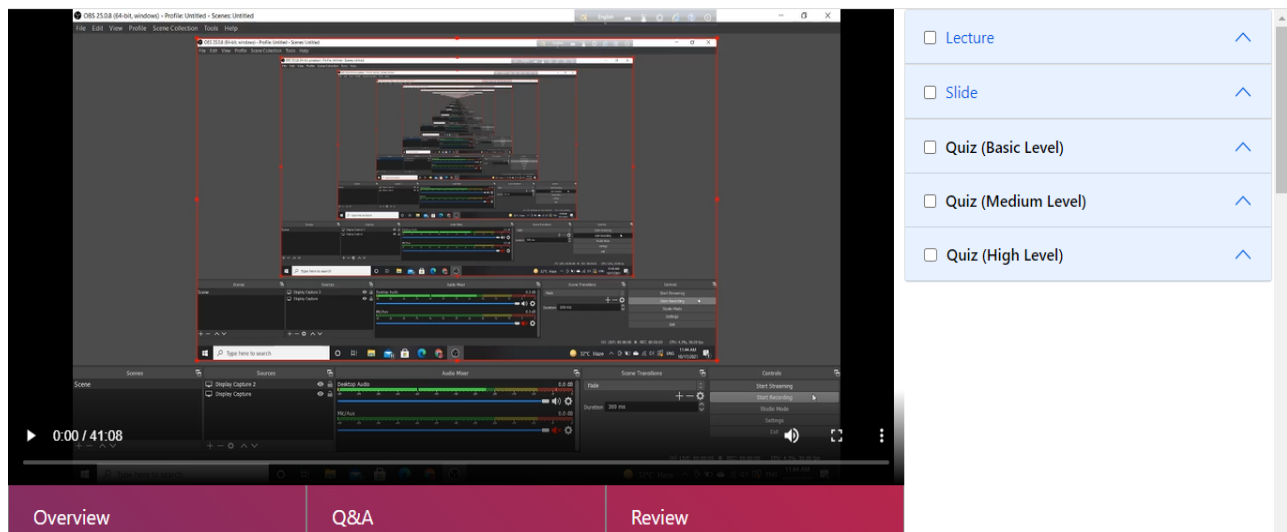


Figure 4.30: Course Content

4.1.24 Course Content Backend

```

public function section($id){
    try{
        $sql = "SELECT * FROM `section` a inner join course s on a.crs_id = s.c_id where crs_id=$id";
        $stmt = $this->db->prepare($sql);
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function vdo($id){
    try{
        $sql = "SELECT * FROM slide where cid=$id";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

```

Figure 4.31: Course Content Backend

4.1.25 QA

Overview

Q&A

Review

Ask Your Question

File Edit View Format

↶ ↷

Paragraph ▾

B *I*

≡ ≡ ≡ ≡

≡ ≡

ASK

Already Asked

shirin

QUESTION : I'm still a bit confused about what was installed during the laravel install

ANSWER : Go to "<https://laravel.com/docs/8.x/installation>"

Add Your Answer

Answer

Figure 4.32: QA

4.1.26 QA Backend

```
public function insertqus($user,$qs,$cid){
    try {
        $sql = "INSERT INTO question (user, ques, cid) VALUES (:user, :ques, :cid)";
        $stmt = $this->db->prepare($sql); $stmt->bindParam(':user',$user); $stmt->bindParam(':ques',$qs); $stmt->bindParam(':cid',$cid);
        $stmt->execute();
        return true;
    } catch (PDOException $e) {
        echo $e->getMessage(); return false;
    }
}

public function qustn($id){
    try{
        $sql = "SELECT id FROM `question` where cid=$id";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage(); return false;
    }
}

public function insertans($id,$ans,$cid){
    try{
        $sql = "UPDATE `question` SET `ans`=:ans WHERE id=:aid AND cid=:cid";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':ans',$ans); $stmt->bindParam(':aid',$id); $stmt->bindParam(':cid',$cid);
        $stmt->execute();
        return true;
    } catch (PDOException $e) {
        echo $e->getMessage(); return false;
    }
}
```

Activate W

Figure 4.33: QA Backend

4.1.27 Course Details

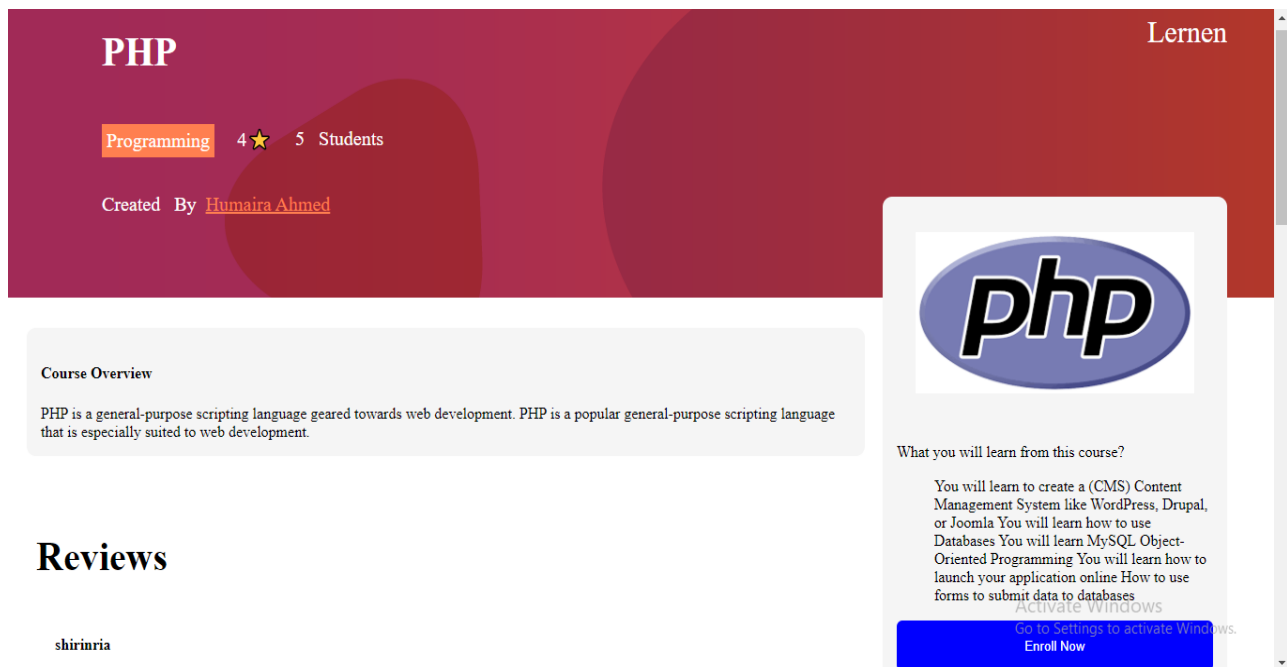


Figure 4.34: Course Details

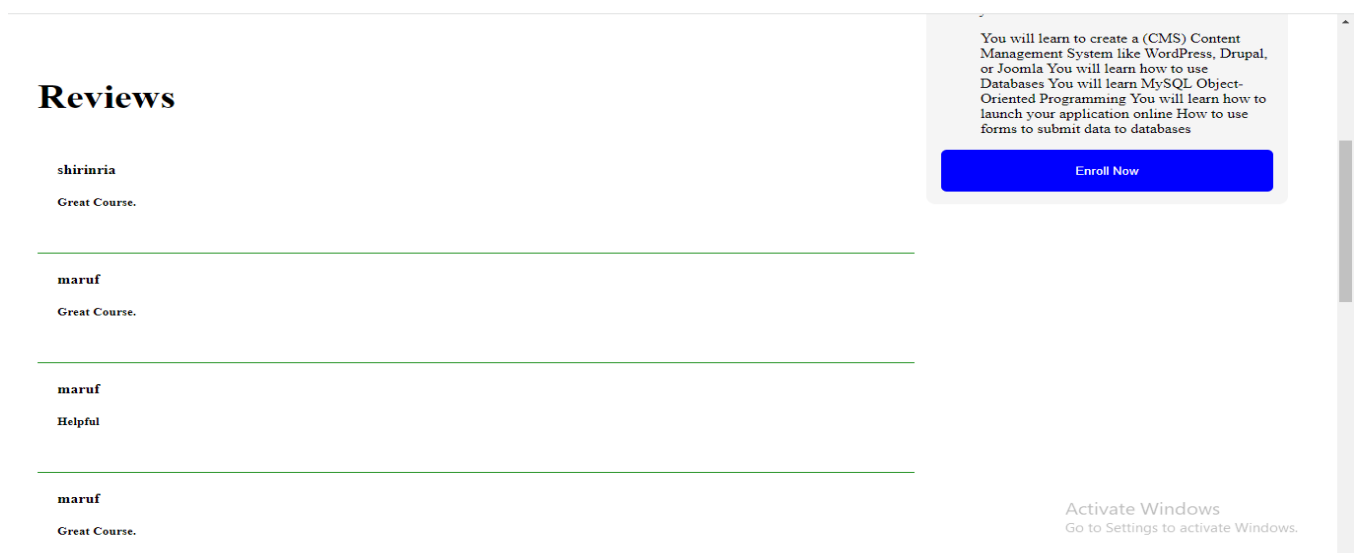


Figure 4.35: Course Details (review section)

4.1.28 Course Details Backend

```
public function enr1($cid){  
    try{  
        $sql = "select * from course where c_id = :id";  
        $stmt = $this->db->prepare($sql);  
        $stmt->bindParam(':id', $cid);  
        $stmt->execute();  
        $result = $stmt->fetch();  
        return $result;  
    }catch (PDOException $e) {  
        echo $e->getMessage();  
        return false;  
    }  
}  
  
public function ecat($ctid){  
    try{  
        $sql = "select * from crs_cat where cat_id = :id";  
        $stmt = $this->db->prepare($sql);  
        $stmt->bindParam(':id', $ctid);  
        $stmt->execute();  
        $result = $stmt->fetch();  
        return $result;  
    }catch (PDOException $e) {  
        echo $e->getMessage();  
        return false;  
    }  
}  
  
public function review($id){  
    try{  
        $sql = "select * from review where cid = $id";  
        $result = $this->db->query($sql);  
        return $result;  
    }catch (PDOException $e) {  
        echo $e->getMessage();  
        return false;  
    }  
}
```

Figure 4.36: Course Details Backend

4.1.29 Course Feedback

Overview

Q&A

Review

Give Feedback

Rating



Write a Review

File Edit View Format

↶ ↷ Paragraph ▾

B *I*

Submit Review

Figure 4.37: Course Feedback

4.1.30 Course Feedback Backend

```
$conn = new mysqli('localhost', 'root', '', 'learnen_two');
$crs_id=$_SESSION['sec'];
$usr_id=$_SESSION['userid'];
if (isset($_POST['save'])) {
    $uID = $conn->real_escape_string($_POST['uID']);
    $ratedIndex = $conn->real_escape_string($_POST['ratedIndex']);
    $ratedIndex++;
    if (!$uID) {
        $conn->query("INSERT INTO stars (rateIndex,user,crs_id) VALUES ('$ratedIndex','$usr_id','$crs_id')");
        $sql = $conn->query("SELECT id FROM stars ORDER BY id DESC LIMIT 1");
        $uData = $sql->fetch_assoc();
        $uID = $uData['id'];
    } else{
        $sql= $conn->query("SELECT * FROM stars WHERE user='$usr_id' AND crs_id='$crs_id'");
        $Data = $sql->fetch_assoc();
        $userR = $sql->num_rows;
        if($userR==0)
            $conn->query("INSERT INTO stars (rateIndex,user,crs_id) VALUES ('$ratedIndex','$usr_id','$crs_id'");
        else
            $conn->query("UPDATE stars SET rateIndex='$ratedIndex' WHERE user='$usr_id' AND crs_id='$crs_id'");
    }
    exit(json_encode(array('id' => $uID)));
}

$sql = $conn->query("SELECT id FROM stars WHERE crs_id='$crs_id'");
$numR = $sql->num_rows;
$sql = $conn->query("SELECT SUM(rateIndex) AS total FROM stars WHERE crs_id='$crs_id'");
$rData = $sql->fetch_array();
$total = $rData['total'];
$avg = $total / ($numR-1);
$_SESSION['ratng']=$avg;
$conn->query("UPDATE course SET rating='$avg' WHERE c_id='$crs_id'");
```

Figure 4.38: Course Feedback Backend

4.2 TEACHER SECTION

4.2.1 Upload Course

Upload Your Content

Course

Book

Status

Exit

Enter Your Course Name

Select Category

Select Your Course thumbnail

No file chosen

Write Course Overview

File Edit View Format

Paragraph

Activate Windows
Go to Settings to activate Windows.

Figure 4.39: Upload Course

Upload Your Content

Course

Book

Status

Exit

Write Course description

File Edit View Format

Paragraph

What will students learn from your course

File Edit View Format

Paragraph

Activate Windows
Go to Settings to activate Windows.

Figure 4.40: Upload Course

4.2.2 Upload Course Backend

```
public function insertcrs($title,$descrip,$destination,$cat,$tname,$tid,$lrn,$cdes){
    try {
        $sql = "INSERT INTO course (c_name,category, des, img_path,tid, tname,learn, cdes )
VALUES (:title, :cat, :descrip, :img,:tid, :tname,:lrn, :cdes)";
        $stmt = $this->db->prepare($sql); $stmt->bindParam(':title',$title);
        $stmt->bindParam(':cat',$cat); $stmt->bindParam(':descrip',$descrip);
        $stmt->bindParam(':img',$destination); $stmt->bindParam(':tid',$tid);
        $stmt->bindParam(':tname',$tname); $stmt->bindParam(':lrn',$lrn);
        $stmt->bindParam(':cdes',$cdes);
        $stmt->execute();
        $last_id = $this->db->lastInsertId();
        return $last_id;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function insertstr( $cid){
    try {
        $sql = "INSERT INTO stars (crs_id ) VALUES (:cid)";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':cid',$cid);
        $stmt->execute();
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.41: Upload Course Backend

```
public function getcrsid($title){
    try{
        $sql = "select * from course where c_id = :title";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':title', $title);
        $stmt->execute();
        $result = $stmt->fetch();
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function insertsec($title,$id,$destination){
    try {
        $sql = "INSERT INTO section (name, crs_id, video) VALUES (:title, :id, :vdo)";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':title', $title); $stmt->bindParam(':id', $id);
        $stmt->bindParam(':vdo', $destination);
        $stmt->execute();
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function catgry(){
    try{
        $sql = "SELECT * FROM `crs_cat` ";
        $result = $this->db->query($sql);
        return $result;
    }catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.42: Upload Course Backend

4.2.3 Upload Books

Upload Your Content

Course

Book

Status

Exit

Enter Your Book Name

Select Category ▼

Enter Author Name

Enter Publisher Name

Edition, e.g: 10th edition

Language

Upload Book's image

Choose file No file chosen

Upload PDF

Choose file No file chosen

Upload

Activate Windows
Go to Settings to activate Windows.

Figure 4.43: Upload Books

4.2.4 Upload Books Backend

```
public function insertbooks( $title,$author,$cat,$pdf,$pub,$ed,$lang,$destination,$size){
    try {
        $sql = "INSERT INTO books (bname,aname,bcat,pdf,pubshr,edtn,lang,img_path,size)
        VALUES (:title, :author, :cat, :pdf, :pub, :ed, :lang, :destination, :size)";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':title',$title);
        $stmt->bindParam(':cat',$cat);
        $stmt->bindParam(':author',$author);
        $stmt->bindParam(':pdf',$pdf);
        $stmt->bindParam(':pub',$pub);
        $stmt->bindParam(':ed',$ed);
        $stmt->bindParam(':destination',$destination);
        $stmt->bindParam(':lang',$lang);
        $stmt->bindParam(':size',$size);
        $stmt->execute();
        return true;
    }
    catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}
```

Figure 4.44: Upload Books Backend

4.2.5 Course Status

The screenshot displays a web application interface for managing course status. On the left is a dark sidebar with navigation links: 'Upload Your Content' (orange), 'Course', 'Book', 'Status' (highlighted in red), and 'Exit'. The main content area has a light gray background. At the top, a gray box shows 'Total Enrollment 7'. Below this is a table with columns: ID, Title, Created Date, and Total Students. The table lists four courses: PHP (ID 1, 5 students), IELTS (ID 2, 0 students), Graphic Design (ID 10, 1 student), and Develop your own game (ID 14, 1 student). Each row has 'Delete' and 'View' buttons. At the bottom right, there is a Windows activation watermark.

ID	Title	Created Date	Total Students
1	PHP	2021-11-03	5
2	IELTS	2021-11-03	0
10	Graphic Design	2021-11-09	1
14	Develop your own game	2021-11-11	1

Figure 4.45: Course Status

4.2.6 Course Status Backend

```
public function status($tid){  
    try{  
        $sql = "select * from course where tid=$tid";  
        $result = $this->db->query($sql);  
        return $result;  
    }catch (PDOException $e) {  
        echo $e->getMessage();  
        return false;  
    }  
}
```

Figure 4.46: Course Status Backend

4.2.7 Update Course Content

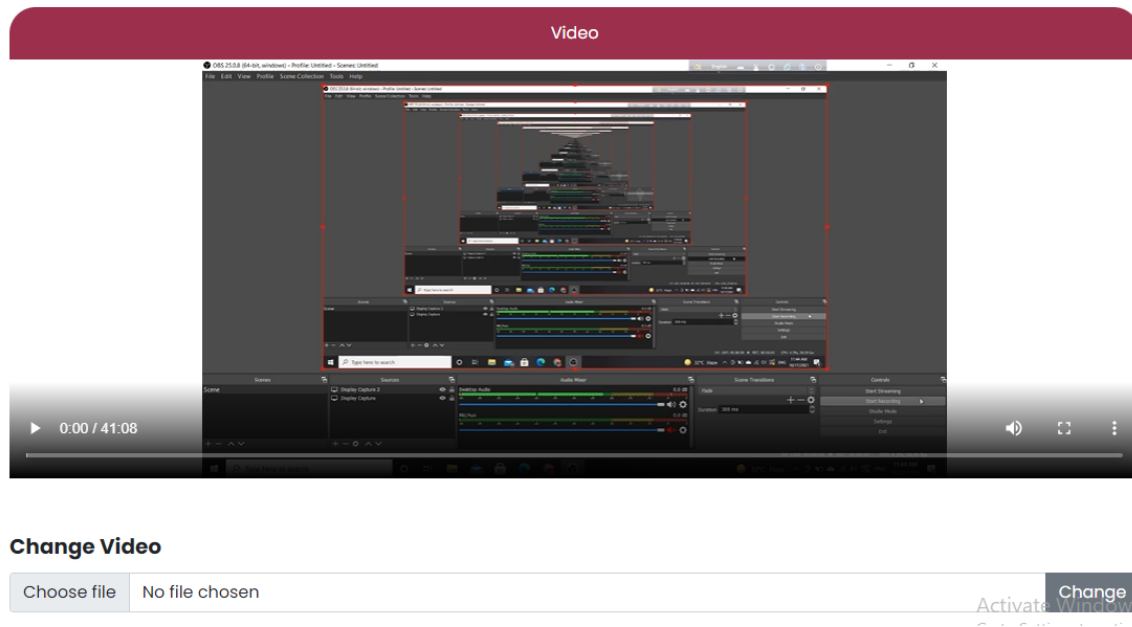


Figure 4.47: Update Course Content

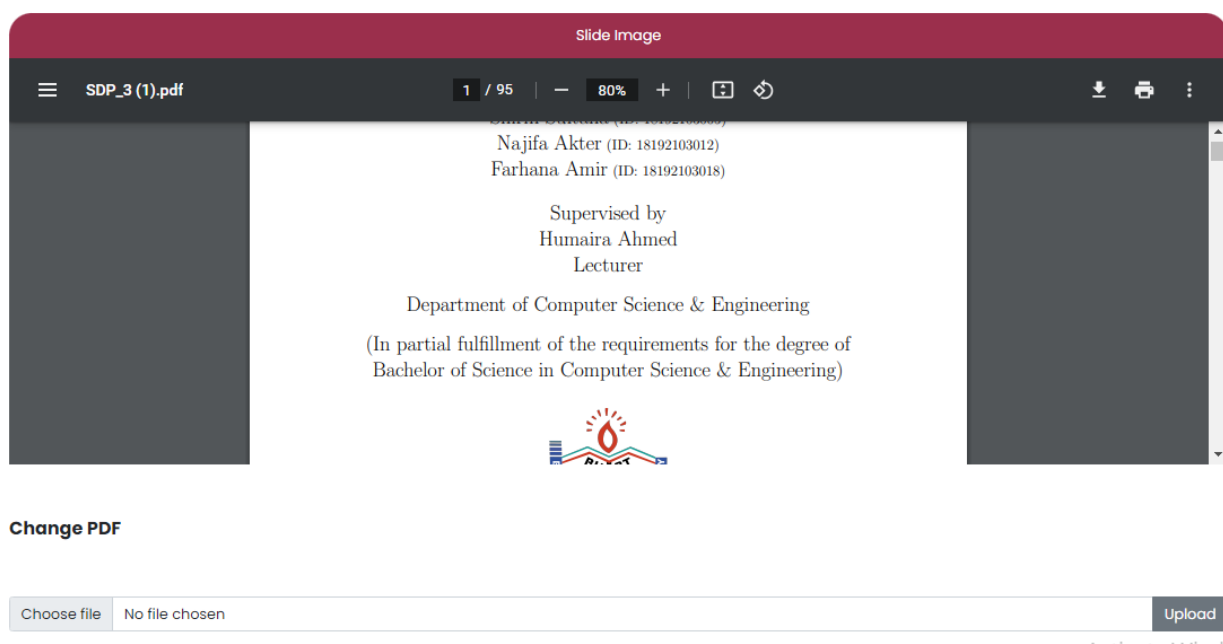


Figure 4.48: Update Course Content

Quiz Questions			
Level	Question	Options	
Basic	What does PHP stand for?	• Personal Hypertext Processor • Private Home Page • PHP: Hypertext Preprocessor Correct Answer: PHP: Hypertext Preprocessor	Delete
Basic	How do you write "Hello World" in PHP	• echo "Helow World"; • "Hello World"; • Document.Write("Hello World"); Correct Answer: echo "Helow World";	Delete
Basic	All variables in PHP start with which symbol?	• \$ • % • ? • ! Correct Answer: \$	Delete
Basic	What is the correct way to end a PHP statement?	• . • New Line • • ; Correct Answer: ;	Delete
Basic	The PHP syntax is most similar to:	• Java • Perl and C • Visual Basic • VB Script Correct Answer: Perl and C	Delete
Medium	When using the POST method, variables are displayed in the URL:	• TRUE • FALSE Correct Answer: FALSE	Delete
Medium	In PHP you can use both single quotes (') and double quotes (") for strings:	• TRUE • FALSE Correct Answer: TRUE	Delete
Medium	What is the correct way to create a function in PHP?	• function myFunction() • new_function myFunction() • create myFunction() Correct Answer: function myFunction()	Delete
Medium	PHP allows you to send emails directly from a script	• TRUE • FALSE Correct Answer: TRUE	Delete
Medium	What is a correct way to add a comment in PHP?	• /*..... */ • Correct Answer: /*..... */	Delete
High	Which of the following is the default file extension of PHP?	• .php • .hphp • .xml • .html Correct Answer: .php	Delete
ADD Questions			

Figure 4.49: Update Course Content

4.2.8 Update Course Content Backend

```
public function viewvdo($id){
    try {
        $sql = "select * from section where crs_id = :id";
        $stmt = $this->db->prepare($sql);
        $stmt->bindParam(':id',$id);
        $stmt->execute();
        $result = $stmt->fetch();
        return $result;
    } catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function viewing($id){
    try{
        $sql = "SELECT * FROM slide where cid=$id";
        $result = $this->db->query($sql);
        return $result;
    } catch (PDOException $e) {
        echo $e->getMessage();
        return false;
    }
}

public function viewcours($id){
```

Figure 4.50: Update Course Content Backend

```
✓ public function viewqus($id){
✓   try{
      $sql = "SELECT * FROM questions where crs_id=$id";
      $result = $this->db->query($sql);
      return $result;
✓ }catch (PDOException $e) {
      echo $e->getMessage(); return false;
    }
  }
✓ public function viewops($id,$qn,$lv){
✓   try{
      $sql = "SELECT * FROM options where crs_id=$id AND question_number=$qn AND level='$lv'";
      $result = $this->db->query($sql);
      return $result;
✓ }catch (PDOException $e) {
      echo $e->getMessage(); return false;
    }
  }
✓ public function viewcrct($id,$qn,$lv){
✓   try {
      $sql = "select * from options where crs_id = :id AND question_number=:qn AND is_correct=1 AND level=:lv";
      $stmt = $this->db->prepare($sql);
      $stmt->bindParam(':id',$id); $stmt->bindParam(':lv',$lv); $stmt->bindParam(':qn',$qn);
      $stmt->execute();
      $result = $stmt->fetch();
      return $result;
✓ } catch (PDOException $e) {
      echo $e->getMessage(); return false;
    }
  }
}
```

Activate Window
Go to Settings to act

Figure 4.51: Update Course Content Backend

Activate Wi-Fi
Go to Settings

Figure 4.52: Update Course Content Backend

4.3 Conclusions

The main aim and focus of this study is to find out what will help in the creation and designing of an e-Learning service called LERNEN. The specific objectives of this study include the following:-

Firstly to investigate the challenges the students faces with the current system.

Secondly is to find out the requirements of the students that will be needed in the new system and how will it meet with the needs of the students.

Thirdly is to find out whether the new system will benefit the students more than the current system.

Fourth is to make the new system user friendly to the students, teachers and others.

Chapter 5

TESTING

5.1 Testing

Testing is the process in which the system is run on manually created input so that the system is correctly working as desired or not.

During systems testing, the system is used experimentally to ensure that the software does not fail. In other words, we can say that it will run according to its specifications and in the way users expect. Special test data are input for processing, and the results examined.

A limited number of users may be allowed to use the system so that analyst can see whether they try to use it in unforeseen ways. It is desirable to discover any surprises before the organization implements the system and depends on it.

Testing of a system is generally done in two phases – one is Unit Testing which is done for each module independently on its completion and the other one is System Testing which is done at the end of a project.

5.2 Validation Criteria

The validation criteria in this project are as follows.

- Here, the user inputs are validated before storing them, and then further for displaying etc.
- All the screens have a similar look and feel. They all have the almost same color combinations in its background. This provides a better user interface to the users.
- The primary key values cannot be duplicated.

5.3 Important Of Testing

During systems testing, the system is used experimentally to ensure that the software does not fail. In other words, we can say that it will run according to its specifications and in the way users expect. Special test data are input for processing, and the results examined.

The importance of system testing is that the system is expected to run according to member's requirement before delivering it to the customer. The System is tested on the basis of specification so that it does not fail on user site.

5.4 Website Administration Functions

The user accessed the website login page by clicking on the Login button at the top right-hand-corner of the main website. This took the user to the Login page, where the user was prompted for an email and a password. This was the expected output of the test case.

5.5 Logging In

The user entered the username and password, which were assigned in the database in the time of registration. The webpage refreshed and displayed the main webpage. This was the expected output of the test case.

5.6 Structural Testing

Structural Testing takes into account the internal mechanism of a system or component. Fatigue Testing is carried out with the objective of determining the relationship between the stress range and the number of times it can be applied before causing failure. So, when our product's structural durability needs to be predicted, verified and validated, turn to DTB's Structural Testing and Fatigue Testing experts.

5.7 Functional Testing

It is very useful and convenient in support of functional testing. Although JMeter is known more as a performance testing tool, functional testing elements can be integrated within the Test Plan, which was originally designed to support load testing. Many other load-testing tools provide little or none of this feature, restricting themselves to performance-testing purposes. Besides integrating functional-testing elements along with load-testing elements in the Test Plan, we can also create a Test Plan that runs these exclusively. In other words, aside from creating a Load Test Plan, it also allows us to create a Functional Test Plan. This flexibility is certainly resource efficient for the testing project.

5.8 Acceptance Test Generation

The objective of this step is to produce a set of test data that may be used to test the system. Whenever a new system is developed it need to be tested to confirm its validity and to determine whether it meets the user requirements. The system was also tested with some sample records. The records were entered into the system and various reports were generated to check the system.

System testing is a critical phase of implementation. Testing of the system involves hardware devices and debugging of computer programs and testing information processing procedures. Testing can be done with test data, which attempt to simulate all possible condition that may rise during processing. The testing methods adopted during the testing of system are unit testing and integration testing.

5.9 Unit Testing

Unit testing focuses on the modules independently to locate the errors. This enables the tester to detect errors in coding. It is the process of taking a module and running it in isolation from rest of the software product by using prepared test cases and comparing the actual result with the result redirected with the specifications and design of the module. One purpose of testing is to find and remove as many errors in the software as practical. There are number of reasons in support of unit testing:

- The size of module single module is small that we can locate an error fairly easily.
- The module is small enough that we can attempt to test it in some demonstrably exhaustive fashion.
- Confusing interactions of multiple errors in widely different parts of software are eliminated.

There is problem associated with testing a module in isolation. How do we run a module without anything to call it, to be called by it, possibly to output intermediate values obtained

during execution. One approach is to construct an appropriate driver routine to call it, and simply stubs to be called by it, and to insert output statements in it. Stubs serve to replace modules that are subordinate to the module to be tested. A stub or dummy subprogram uses the subordinate module's interface, may do minimal data manipulation, prints verification of entry and returns.

5.10 Integration Testing

This is a systematic technique for constructing the program structure while at the same time to uncover the errors associated with the interface. The objective is to take unit tested module and build a program structure that has been detected by designing. The main purpose of integration testing is to determine that the interfaces between modules are correct or not. One specific target of integration testing is the interface: whether parameter matches on both sides as to type, permissible ranges, meaning utilization. There are 3 types of integration testing-

- **Top-Down Approach :** Top-Down integration proceeds down the invocation hierarchy, adding one module at a time until an entire tree level is generated.
- **Bottom-Up Approach :** The Bottom-up strategy works similarly from the bottom to up.
- **Sandwich Strategy :** A sandwich strategy runs from top and bottom simultaneously.

5.11 Test Data Used

The proper selection of the data is very important. If the test data is not appropriate or representative of the data to be provided by the user, the reliability of the output is susceptible.

Two different sources were during testing of the system:

- **Using Live Test Data :** Live test are those that are actually extracted from the organization files. Use of the live data make testing easier by obtaining most expected outputs and if it is found that the program can handle the entries processing of the system accurately.
- **Using Artificial Test Data :** Live data is difficult to obtain insufficient amount to conduct extensive testing. It does not test all the combination or formats that can be done by entering to the system. Therefore, artificial test data were used at the time of unit testing. Artificial test data was created solely for test purposes which provide extreme values for testing the limit of candidate system.

5.12 Test Cases

- **System is properly linked or not :** Whether they are redirected to desired page or not.
- **Information passed :** If a page passes some parameter to another page, then it should be checked that the page gets the correct information, whatever is passed by the previous page.
- **Output should be correct :** Every functionality of the system should be checked properly whether it gives the right result or not generally test is performed with known results. If the output of the system is matched with that result the system is working fine.

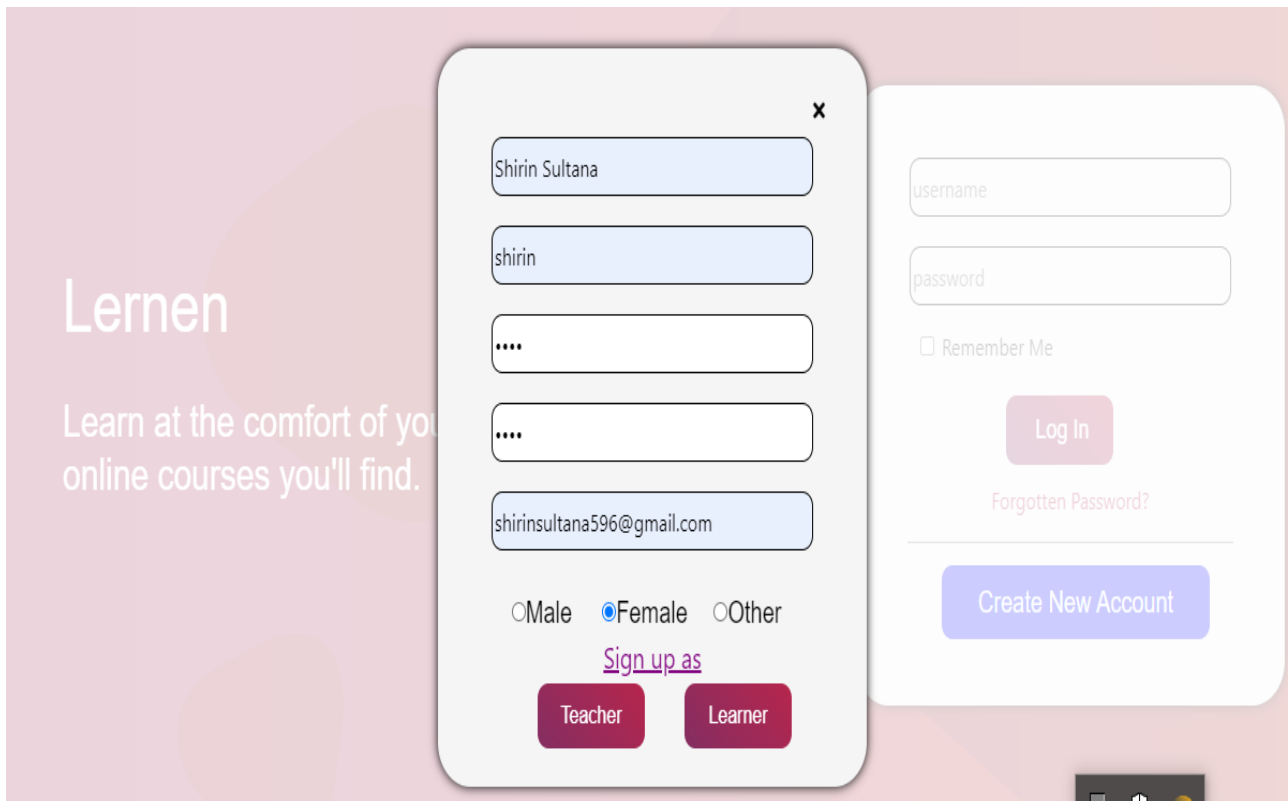
Module	Test Case Scenarios	Tested	Working
Index	1. Verifying design, style fonts etc. 2. Verifying link /external link 3. Verifying all buttons and key	Yes	Yes
Register	1. Checking database connection and UI design 2. Verify all textbox are working and valid 3. Checking duplication and Adding data in database 4. Checking if OTP was sent via email 5. Checking Email and Password authentication	Yes	Yes
Login	1. Checking database connection and UI design 2. Verifying Username and pass form and validity 3. Checking Email, Username and pass authentication 4. Checking the COOKIE setting when marked the Remember me box	Yes	Yes
Account	1. Checking All information are retrieved from database 2. . Validating form and update data	Yes	Yes
Course Page	1. Checking database connection 2. Checking UI of course loading design 3. Retrieve course from database 4. Verify all function are working	Yes	Yes

Course content Page	1. Checking database connection 2. Checking UI of course content loading design 3. Retrieve course content from database 4. Add questions and answers of the Q/A section in the database and retrieve 5. Add and retrieve feedback 6. Verify all function are working	Yes	Yes
Search Page	1. Verify database functionality 2. Checking UI design 3. Whether search working 4. Verify all searching functionality	Yes	Yes
Book Page	1. Checking database connection 2. Checking UI of Books loading design 3. Retrieve PDF from database 4. Verify the download functionality	Yes	Yes
Upload Course/ Book	1. Checking database connection and UI design 2. Add Course video , thumbnail, category in database 3. Whether quiz questions with options and levels are inserted in the database according to course 4. Add book's PDF , thumbnail, category in database 5. Checking the upload course/books functionality	Yes	Yes
Update	1. Checking whether the changed content of a course update in the database according to that course 2. Whether quiz questions with options and levels are inserted or deleted in the database	Yes	Yes

Table 5.1: Testing Cases for Lernen

5.13 Testing result

Sign in system work perfectly.If given password doesn't match with database then it show warning.If given user mail doesn't match with database then also show warning.So we can say here sign in system work perfectly.



The image shows a web application interface with a sign-up modal and a login form. The background is a light pink color with the word "Lernen" and the text "Learn at the comfort of your own pace. Find the online courses you'll find." in a light pink font. The sign-up modal is a white box with rounded corners and a close button (X) in the top right corner. It contains the following fields and options:

- First Name: "Shirin Sultana"
- Last Name: "shirin"
- Phone Number: "...."
- Address: "...."
- Email: "shirinsultana596@gmail.com"
- Gender: Radio buttons for "Male", "Female" (selected), and "Other".
- Sign up as: A link "Sign up as" in purple text.
- Role: Two buttons, "Teacher" and "Learner", in maroon color.

The login form is a white box with rounded corners, positioned to the right of the sign-up modal. It contains the following fields and options:

- Username: "username"
- Password: "password"
- Remember Me: A checkbox and the text "Remember Me".
- Log In: A button in maroon color.
- Forgotten Password?: A link in light pink text.
- Create New Account: A button in light blue color.

Figure 5.1: Sign Up

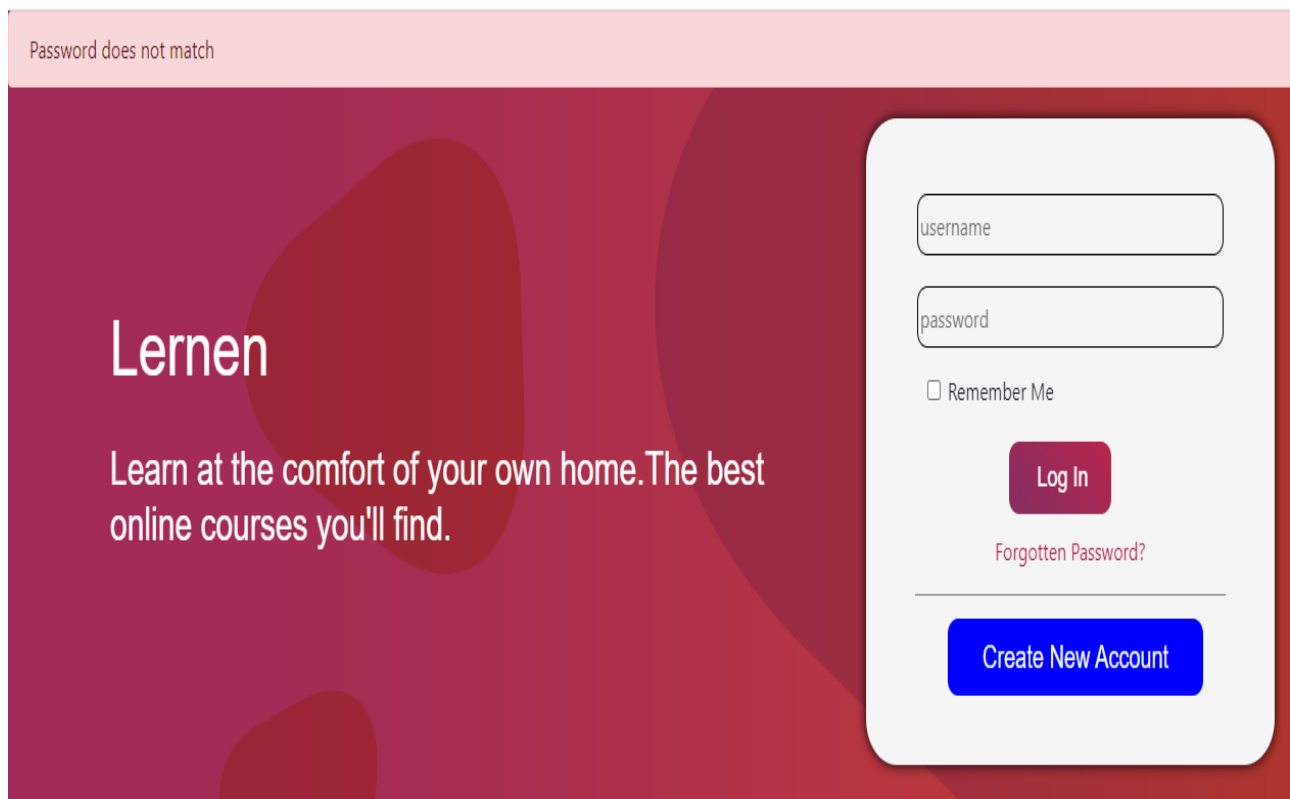


Figure 5.2: warning show for wrong password

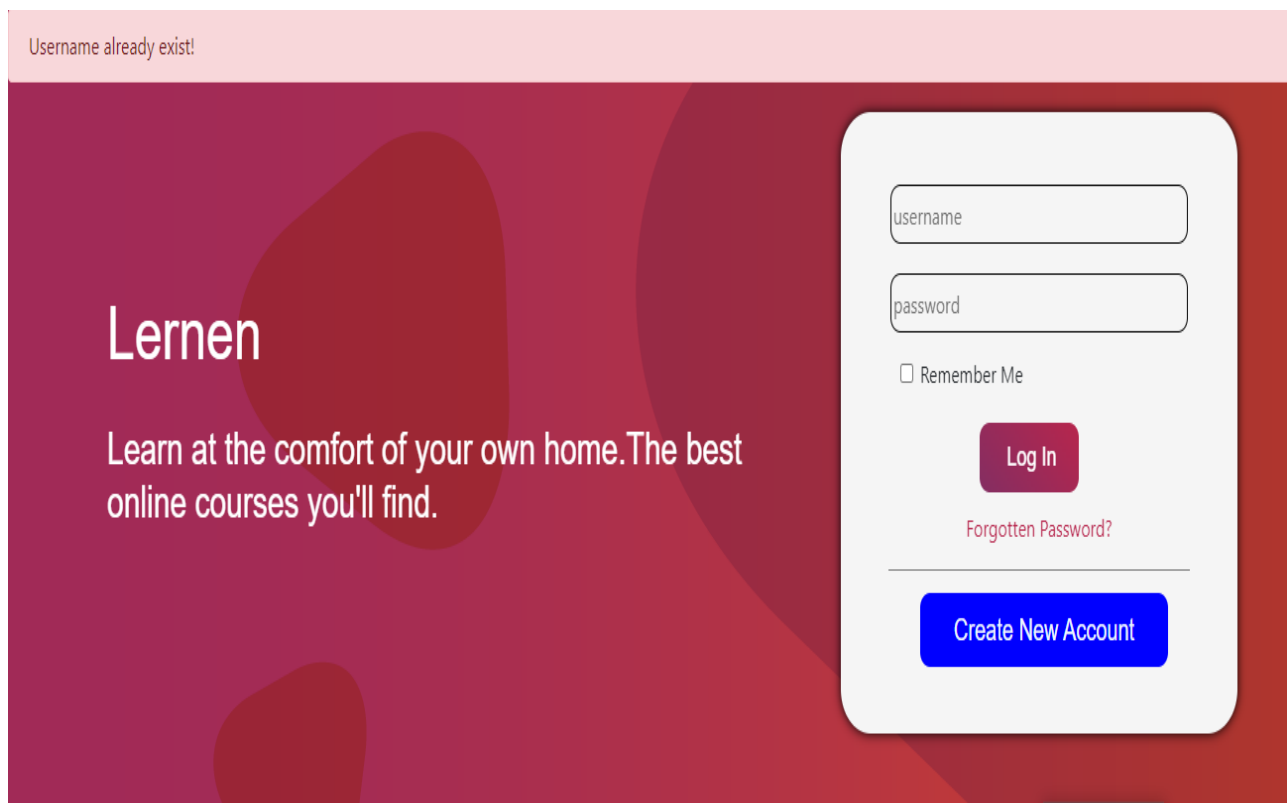


Figure 5.3: warning show for wrong username

Then we tested upload course and upload books page. We input dummy value and that works correctly. Then we fill up other boxes and we upload image, PDF and video that also works correctly. Then click upload button and then successfully create the course or book. The data also successfully stored in database. We also added questions for quiz test. They also added successfully.

Upload Your Content

Course

Book

Status

Exit

PHP

Programming

Select Your Course thumbnail

Choose file

PHP.jpg

Write Course Overview

File Edit View Format

↶ ↷ Paragraph B I

PHP is a general-purpose scripting language geared towards web development. PHP is a popular general-purpose scripting language that is especially suited to web development.

Write Course description

Write Course description

File Edit View Format

↶ ↷ Paragraph B I

Are you new to PHP or need a refresher? Then this course will help you get all the fundamentals of Procedural PHP, Object Oriented PHP, MySQLi, and end the course by building a CMS system similar to WordPress, Joomla, or Drupal.

What will students learn from your course

File Edit View Format

↶ ↷ Paragraph B I

You will learn to create a (CMS) Content Management System like WordPress, Drupal, or Joomla

You will learn how to use Databases

Next

Figure 5.4: Upload Course



Figure 5.5: Upload Course Materials

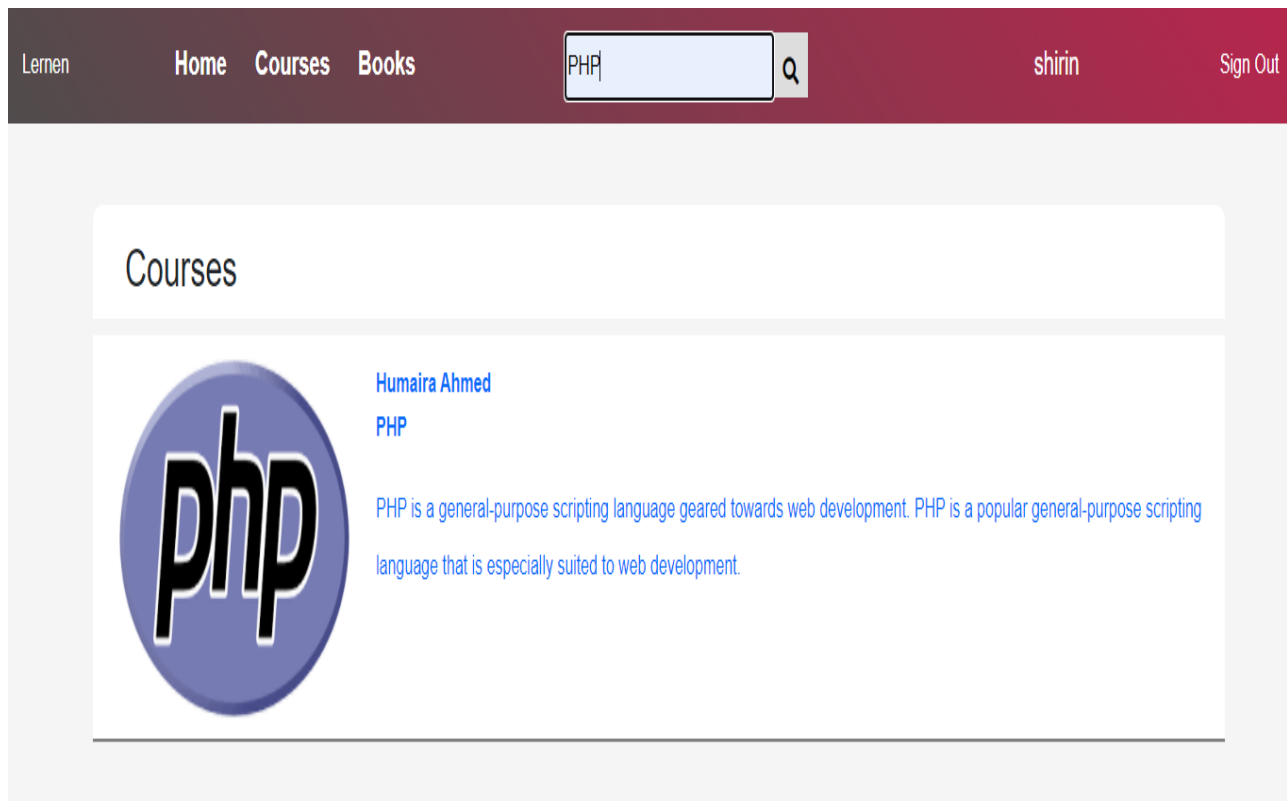


Figure 5.6: Course Created successfully

				category	c_id	c_name	Date	des	img_path	tid	tname	cdes	learn	rating
☐				13	1	PHP	2021-11-03	PHP is a general-purpose scripting language geared...	uploads/PHP.jpg	2	Humaira Ahmed	Are you new to PHP or need a refresher? Then this ...	You will learn to create a (CMS) Content Managemen...	3
☐				12	2	IELTS	2021-11-03	The International English Language Testing System,...	uploads/IELTS.png	9	Shirin Sultana		NULL	1
☐				2	4	Web Design	2021-11-03	Web design encompasses many different skills and d...	uploads/Web Design.jpg	9	Shirin Sultana		NULL	0
☐				2	10	Graphic Design	2021-11-09	Graphic design is the profession and academic disc...	uploads/Graphic Design.jpg	2	Humaira Ahmed		NULL	2
☐				1	14	Develop your own game	2021-11-11	Sometimes game development is a challenging, painf...	uploads/Develop your own game.jpg	2	Humaira Ahmed	Game Development is the art of creating games and ...	Master beginner C# concepts, like variables, "if" ...	0
☐				1	15	Frontend web development Course	2021-11-11	We'll Cover HTML5, CSS3, JavaScript, jQuery, Boots...	uploads/Frontend web development Course.jpg	9	Shirin Sultana	Web development is the work involved in developing...	You'll learn all the fundamentals of Front-End Web...	3.3333333333333333

Figure 5.7: successfully stored in database

aname	bcat	bid	bname	pdf	publshr	edtn	size	img_path
Weygandt Kimmel Lieso	Academic(College)	9	Accounting	Accounting Principles, 10th edition.pdf	John Wiley & Sons, Inc	10th edition	28188938	uploads/download (2).jpg
Vishal Layka	Development	10	Learn Java for Web Development	Learn Java for Web Development .pdf			NULL	uploads/java.jpg
Noel Kalicharan	Programming	14	Learn to Program with C	Learn to Program with C_ Learn to Program using th...			3	uploads/c.jpg
V.V.K. Subburaj	GK	15	Basic General Knowledge Book	Basic General Knowledge Book (PDFDrive).pdf	Sura College of	(1 January Competition 2012)	5	uploads/Basic General Knowledge Book.jpg
Monira Khaton	IELTS	16	Best Practice Book for IELTS Writing 230 IELTS Wri...	Best Practice Book for IELTS Writing 230 IELTS Wri...			4	uploads/41i809t4r+L.jpg
Glenn Beck	Islam Shikkha	17	It IS About Islam	It IS About Islam_ Exposing the Truth About ISIS, ...		August 18, 2015	2	uploads/It IS About Islam_ Exposing the Truth Abou...
Rupayon Neogy	Design	18	Web Design & UI_UX	Web Design & UI_UX (PDFDrive).pdf			3	uploads/html.jpg
Andy Ilachinski	Photography	19	BWClassNotesBelnavisGallery	BWClassNotesBelnavisGalleryFeb2011.pdf		Feb2011	6	uploads/1-03c8a2bd48.jpg
Trishna Knowledge Systems	Academic(School)	20	Chemistry (Class 9)	Chemistry (Class 9) (PDFDrive).pdf		2014	33	uploads/382f5ec66911100097c5d3d69b6cb06f

Figure 5.8: successfully stored in database

Add A Question

Question Number:

Question Text:

Choice 1:

Choice 2:

Choice 3:

Choice 4:

Correct Option Number



Figure 5.9: Add Quiz Questions

We also test our update course content section. They were also changed successfully.

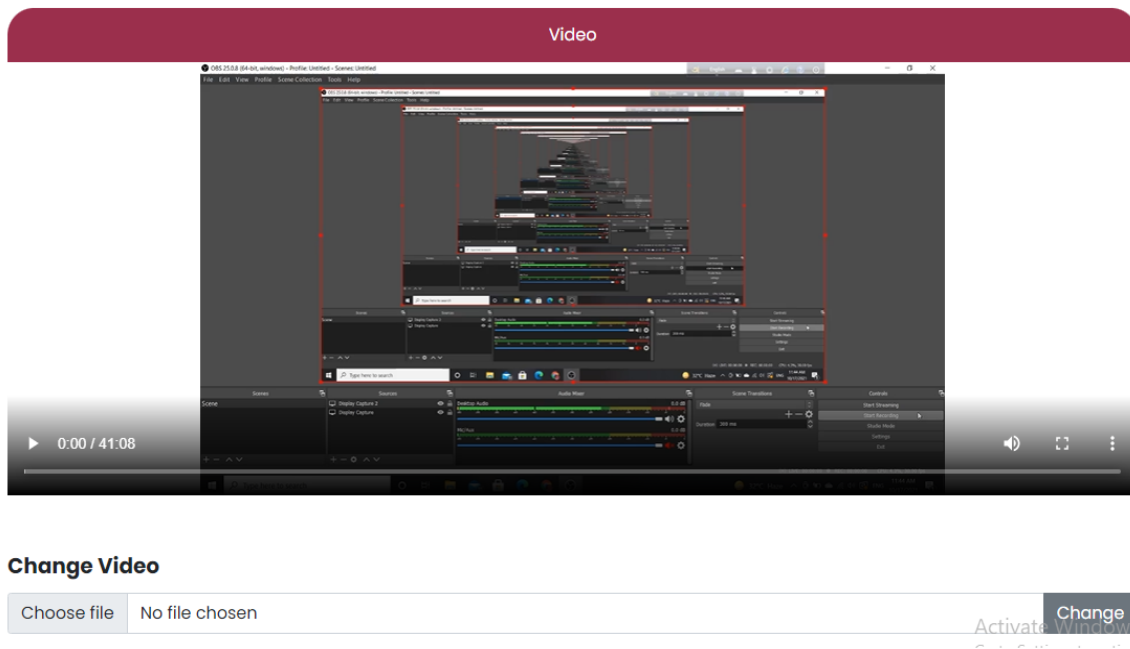


Figure 5.10: Update Course Content

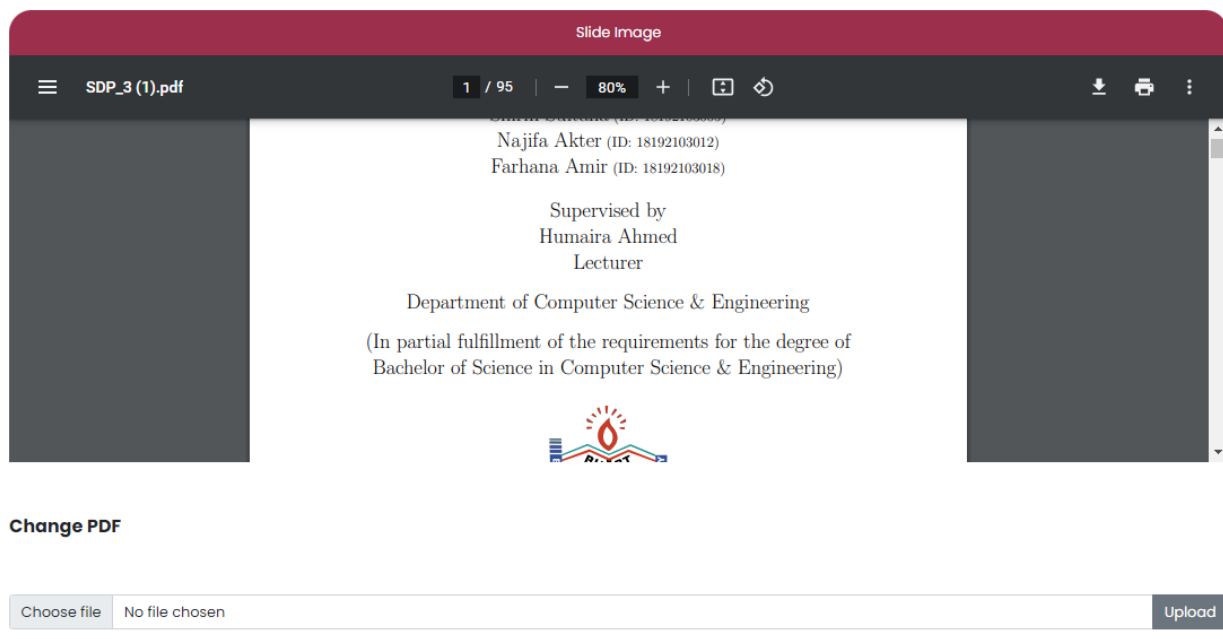


Figure 5.11: Update Course Content

Quiz Questions			
Level	Question	Options	
Basic	What does PHP stand for?	• Personal Hypertext Processor • Private Home Page • PHP: Hypertext Preprocessor Correct Answer: PHP: Hypertext Preprocessor	Delete
Basic	How do you write "Hello World" in PHP	• echo "Helow World"; • "Hello World"; • Document.Write("Hello World"); Correct Answer: echo "Helow World";	Delete
Basic	All variables in PHP start with which symbol?	• \$ • % • ? • ! Correct Answer: \$	Delete
Basic	What is the correct way to end a PHP statement?	• . • New Line • • ; Correct Answer: ;	Delete
Basic	The PHP syntax is most similar to:	• Java • Perl and C • Visual Basic • VB Script Correct Answer: Perl and C	Delete
Medium	When using the POST method, variables are displayed in the URL:	• TRUE • FALSE Correct Answer: FALSE	Delete
Medium	In PHP you can use both single quotes (') and double quotes (") for strings:	• TRUE • FALSE Correct Answer: TRUE	Delete
Medium	What is the correct way to create a function in PHP?	• function myFunction() • new_function myFunction() • create myFunction() Correct Answer: function myFunction()	Delete
Medium	PHP allows you to send emails directly from a script	• TRUE • FALSE Correct Answer: TRUE	Delete
Medium	What is a correct way to add a comment in PHP?	• /*..... */ • Correct Answer: /*..... */	Delete
High	Which of the following is the default file extension of PHP?	• .php • .hphp • .xml • .html Correct Answer: .php	Delete
ADD Questions			

Figure 5.12: Update Course Content

5.14 Testing Summery

We tested all the panels assigned for users. The project was tested in the localhost with all possible options. The testing involved using some dummy records and inserting them to populate the database and to check whether the features are producing results properly.

The testing process involved:

- Testing for the successful login process.
- Testing for form validation.
- Testing for the email sending and validation process.
- Testing for the course enroll process.
- Testing for the book download process.
- Testing for proper record generation and maintenance.

The project was successfully tested and few bugs were fixed. It was discovered during the testing that some features would be easier to implement in some alternative manner, and their implementation methods were changed.

Chapter 6

USER MANUAL

6.1 Introduction

The Learning Management System provides the capability to manage and administer all aspects of your training program by providing a complete training and testing system to fully address your knowledge requirements. By creating powerful training resources and delivering such resources online; supporting both online training and traditional classroom training; providing online testing and scoring of each participant's performance; and real-time reporting to managers and training personnel, the Learning Management System provides a powerful, flexible and scalable system which is cost effective and easy to use.

To full-fill our project or to build our website we had some hardware and software requirements. In this chapter we will describe about what we needed to complete this work.

6.2 System Requirements

6.2.1 H/W Requirements

- **Operating System** : Windows 98, Windows XP, Windows 7, Linux
- **Processor** : Dual core processor with a base clock speed above 1.5GHz
- **64-bit Computing** - Yes
- **Ram** : minimum 2GB
- **Storage** : 128GB
- **Bandwidth** : Minimum of 512 kbps.
- **Disk space** : 160MB free (min) plus as much as you need to store materials. 5GB is probably a realistic minimum.

6.2.2 S/W Requirements

- **Operating System** : Windows 7/ XP/8/10
- **Front End** : HTML, CSS, Java Script.
- **Front End Framework** : Bootstrap 5
- **Server Side Script** : PHP
- **Database** : MySQL
- **Local server setup package** : XAMPP,
- **Dependency Manager** : Composer
- **Browser** : Browser will be used to check the system visuals and test the functionalities while developing. Any of Mozilla Firefox, Opera, Chrome etc. can be used. But Chrome is recommended.

6.3 User Interface

The very first portion is “User Login” which is known as the user interface. As this is authorized to access, the user needs to validate the user name and password. If the user forgets the password, he can recover his password using forget password function. Then there will be a “Home Page” where users can see newly launched courses and Trending courses. If the user taps on the courses, all the courses on the website will be shown. This page is known as the “All Course List” page. If the user wants to see any specific category’s course, he has to tap on that category of course. Users can search courses from the search bar. The searched keyword will be matched with the course name and description. Now if the user wants to enroll or see the rating, review of any particular course, then he will tap on that course. This is known page as enroll page. To enroll he will tap on enroll button. To give feedback on any particular course, then he will tap on review on vdo page. To ask any question tap on the QA button. For more details about the course and instructor see the overview section. To give the quiz test, tap on the quiz button. This will open a new tab containing specific leveled questions. After the test, the user can see the result and correct answer. To check registered course status tap on the history button which is under the profile button. The user’s registered course will be stored on my course page. To see and update personal information go to the profile page.

To upload courses tap on the upload button which is beside the profile. From there, instructors can upload course materials and PDF.

The instructor can also check course status. To change or add any content of course tap on the view button on the status page.

To download any books go to the books section and press the download button and to see details tap view details. Then users can see the book’s specifications, can download a book, or read a book without download.

There is also the guest section on our website. Guests can see course overview, rating, review. To enroll in a course they have to register first. They can download or read books without registration.

We use cookies to remember users. If the user checks the remember me box. They haven't logged in again. They can enter into the website without login at any time.

6.4 Conclusion

In this chapter, we describe the requirement specifications of the system and the actions that can be done on this things. We also show and explain the GUI (Graphical User Interface) of the system. In conclusion, to this chapter, user will get knowledge about the full working system. This is not a difficult task. User will discover the basic characteristics and the way to use them. They may also know the computer configuration requirements for using this software.

Chapter 7

CONCLUSION AND FUTURE WORKS

7.1 Conclusion

Our project is only a humble venture to satisfy the needs to manage their project work. Several user friendly coding have also adopted. This package shall prove to be a powerful package in satisfying all the requirements of the students. The objective of software planning is to provide a frame work that enables the manger to make reasonable estimates made within a limited time frame at the beginning of the software project and should be updated regularly as the project progresses.

At the end it is concluded that we have made effort on following points:

- A description of the backend and context of the project and its relation to work already done in the area.
- Make statement of the aims and objectives of the project.
- The description of the purpose,scope and applicability.
- We define the problem on which we are working in the project.
- We describe the requirement specifications of the system and the actions that can be done on these things.

- We understand the problem domain and produce a model of the system ,which describes operations that can be perform by the system.
- We included features and operations in detail, including screen layouts.
- We designed user interface and security issues related to the system.
- Finally the system is implemented and tested according to test cases.

7.2 Outcomes

- Added flexibility and self-paced learning.
- Better time management.
- Demonstrated self-motivation.
- Improved virtual communication and collaboration.
- Unlock more learning opportunities.
- Online classes are more affordable.
- To help network with people from around the globe.
- To make refined critical-thinking skills.
- To earn new technical skills.

7.3 Current Limitations

We have some limitations to overcome in future. Some of them are shared below.

- Improve the system's quality and performance.
- It may be a "solo" act.
- We faced some problems with the session and redirecting pages. So, We have a plan to create a cookie for our project to handle sessions and redirecting pages.
- We want to increase the security level one step ahead. Therefore, we want to enable the second step verification for the users.
- For personal servers, we want to enable the MAC address binding feature for devices. So, that unauthorized device cannot access the server.
- Enable live class system.
- There are few complications on Feedback feature. We want to make it more comfortable for the users.

7.4 Future Works

Our website may be manipulated in diverse vital works and may be carried out without problems if we will attain our goal. Although a lot of works are being done, But the use of e-learning websites with the help of our work can be used for far some better learn that are not done yet with that much efficiency.

- **Security** : In future we want to make this site more secure.
- **Efficient System** : Main destiny scope of our proposed version is to strive it in actual global which means we should enforce it with higher results. We can even try and enhance the safety of consumer identity and to make it greater secure. We can give more advance software for this website including more facilities
- **Reduce Time and Complexity** : We want to enhance the facts passing time and facts processing potential. So that we are able to enhance the facts transaction speed, processing potential and most significantly the safety the usage of our system.
- **Add videos playlist** : Our Website wishes greater required nodes to run better learn for users. We want to add a playlist with every course. Currently there is only one video per course.
- **Developed Database**: Integrate multiple load balancers to distribute the loads of the system. Create the master and slave database structure to reduce the overload of the database queries. Implement the backup mechanism for taking backup of codebase and database on regular basis on different servers.
- **Cloud Server**: A cloud server is powerful physical or virtual infrastructure that performs application- and information-processing storage. It is most often a highly powerful virtual IT infrastructure that can host data, applications, information, and other components. It can also be a physical infrastructure with virtual machines. In future we will try to use cloud server instead of xampp.